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Self-diffusion in polymer systems studied by magnetic field-gradient spin-echo NMR methods

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Nomenclature

General concepts

A_2	second virial coefficient
α, ν, l, λ	scaling parameters
B	magnetic flux density (or "magnetic field strength")
β	stretched exponential exponent
c^*	overlap concentration
c, ϕ	concentration
CMC	critical micelle concentration
D	self-diffusion coefficient
$\langle D \rangle$	average self-diffusion coefficient
Da	unit of molecular mass (or weight)
DDC	distribution of (self-)diffusion coefficients
DLS	dynamic light scattering
δ	duration of magnetic field-gradient pulse
f	friction factor
G	magnetic field-gradient
γ	magnetogyric ratio
I	NMR signal intensity
ILT	inverse Laplace transformation
M	molecular mass (or "weight")
M_e	entanglement molecular mass
M_n	number-average molecular mass
M_w	weight-average molecular mass
MRI	magnetic resonance imaging
MSD	mean-square displacement
MWD	molecular weight (or mass) distribution
N	number of statistical segments in a polymer chain
NMR	nuclear magnetic resonance
PGSE	pulsed field-gradient spin-echo
p	degree of binding
$p(D)$, $p(M)$	distribution functions of self-diffusion coefficients and molecular masses, respectively
Π	osmotic pressure
RF	radio-frequency
Z	root-mean-square displacement
r_H	hydrodynamic radius
R_g	radius of gyration
SANS	small-angle neutron scattering
σ	electrolyte conductivity
T	unit of Tesla
T	temperature
T_g	glass-transition temperature
T_m	melting point temperature
T_1 and T_2	spin–lattice and spin–spin relaxation time, respectively
t_d	self-diffusion observation time
τ	time separation between the RF pulses in a Hahn spin-echo experiment
ZAC	zero-average contrast
ξ	mesh size of polymer network

Chemicals and polymers

AIBN	azobisisobutyronitrile
BMA	butyl methacrylate
Boc-Gly	*tert*-butyloxycarbonyl-glycine
Boc-Phe	*tert*-butyloxycarbonyl-L-phenylalanine
CH	cyclohexane
CTAB	hexadecyl trimethylammonium bromide
DMF	dimethylformamide
DMSO	dimethylsulfoxide
EHEC	ethyl (hydroxyethyl) cellulose
EO	ethylene oxide
EPDM	ethylene-propylene-diene
MAA	methacrylic acid
NIPA	*N*-isopropylacrylamide
PB	polybutadiene
PE	poly(ethylene)
PI	polyisoprene
PP	polypropylene
PS	polystyrene
PEB	poly(ethene,butylene)
PEG	poly(ethylene glycol)
PEO	poly(ethylene oxide)
PEP	poly(ethylene-*co*-propylene)
PPO	poly(propylene oxide)
PVE	poly(vinylethylene)
PVF	poly(4-vinylphenol)
PVP	poly(4-vinylpyridine)
didodecyl-PPE	poly((2,5-didodecyl-*p*-phenylene)ethynylene)
didodecyl-PPX	poly(2,5-didodecyl-*p*-xylylene)
PDMA	poly(2-(*N*,*N*-dimethylamino)ethyl methacrylate)
PDMS	poly(dimethylsiloxane)
PMMA	poly(methyl metacrylate)
PNIPAM	poly(*N*-isopropylacrylamide)
SDS	sodium dodecyl sulfate
THF	tetrahydrofuran
TOPO	trioctylphosphine oxide

1. Introduction

1.1. Polymers and polymer systems

Polymers are macromolecules that consist of repeating structural units, called monomers, connected by covalent chemical bonds. Homopolymers consist of only one type of repeating unit, while copolymers consist of two or more different monomers. To the first group of polymers belong polyethylene, polystyrene and polyisoprene (rubber). To the second group belong examples such as nylon, and natural polymers like proteins and polysaccharides. Copolymers may have the different monomers spread randomly along the chains, forming random copolymers, or grouped together in block structures, forming so-called block copolymers. The molecular weight of polymer molecules may be in the range a few hundreds to millions of Daltons (Da). The polymer chains may be linear, branched, or even cross-linked. In addition, side-groups that differ in chemical composition from the monomers in the main chain may be chemically linked to the polymer chain, forming what is known as grafted polymers. Naturally produced polymers, like proteins, are usually monodisperse, i.e. are characterized by one single molecular weight. However, for synthetically made polymers there will always be a certain distribution of molecular weights in a given sample. The degree of polydispersity is dependent on the synthetic procedure, and certain polymers may be made in the laboratory with only a minor degree of molecular weight polydispersity. Some polymers may crystallize at a temperature below a characteristic polymer dependent crystallization, or melting, temperature, T_m. Above this temperature, they exist as a melt, i.e. a liquid of pure polymer. At a certain temperature below T_m, there exists a characteristic temperature called the glass-transition temperature, T_g, with, usually, $0.5 < T_g/T_m < 0.8$ [1]. When polymers that crystallize are below their T_g, they behave as a glass. Between their T_g and T_m, this kind of polymer may have the right mechanical properties for use as a material.

Polymers may be dissolved in solvents of low molecular weight, forming polymer solutions. For the solution to be thermodynamically stable, the molecular interactions between the solvent molecules and the polymer units must be favorable, so that the units are solvated. This stability is also manifested through the second virial coefficient for the solution, in that the coefficient must be sufficiently large and positive for the solution to be stable. In that case, the solvent is termed a good solvent for the actual polymer. In the other extreme, when the second virial coefficient is negative for a certain combination of polymer and solvent, the solvent is termed a bad solvent. In this case, the solution is not thermodynamically stable, and the polymer will separate from the solvent. There is also a border-case, when the combination of polymer and solvent makes the second virial coefficient equal to zero. The solvent is then termed a theta solvent.

For block copolymers one usually has the problem that whatever the solvent, it may only be a good solvent for one of the blocks. This circumstance may, however, be turned into an advantage, in that the blocks for which the solvent is a poor one may form intermolecular physical associations. This may lead to polymer network formation, which is important in e.g. gel systems, that belong to the type of systems generally known as soft materials. The thermodynamics in such systems, as measured by the second virial coefficient, is generally temperature dependent.

General texts on polymer systems and polymer solutions may be found in Refs. [1–5]. Doi and Edwards have written a text on the theory of polymer dynamics [6]. The dynamics in polymer systems may be studied using many different techniques. NMR techniques may be used to determine the self-diffusion of the various components in such systems, giving detailed information on dynamics and even on structure at the molecular level; this review is focused on the latter approach.

1.2. Pulsed field-gradient NMR

The pulsed field-gradient spin-echo (PGSE) NMR method dates back to the year 1965 [7]. In the 45 years that have elapsed since this pioneering work, a large number of publications have dealt with the study of molecular self-diffusion in systems of various complexity. In 1968 Vold and coworkers [8] suggested Fourier transformation of the (second half) of the spin-echo to extract the individual diffusion coefficients in multicomponent liquid systems from partly relaxed spectra. In the most basic form, the PGSE experiment is based on the 90°–τ–180° spin-echo radio-frequency (RF) sequence used in, for example, the Carr–Purcell method for spin–spin (T_2) relaxation measurements [9] to which magnetic field-gradients have been added. At a time τ after the 180° pulse (usually of the order of 10–100 ms) a spin-echo is formed. If the Fourier transformation starts at this point, the resulting spectrum will be essentially identical to the one obtained using an ordinary 90° RF pulse sequence, except for the attenuation of the various signals, brought about by the spin–spin relaxation over the time span of 2τ. This attenuation is in principle given by $\exp(-2\tau/T_2)$. The values of T_2 are different from one component to the other, and also in principle from one NMR signal to another in the same component. Therefore, there will always be a restriction on the range of τ that may be used in such experiments.

Magnetic field-gradients must be absent for the Fourier transformation of the second half of the echo to yield a high-resolution spectrum. In practice, the magnetic field (B) will not be completely homogenous over the sample, but during acquisition of the echo time signal the conditions in the probe have to correspond to those needed for producing high-resolution spectra. This means that one can not observe spin-echoes intended for Fourier transformation if a steady gradient of the magnitude needed for the study of molecular self-diffusion in liquids, is present during the acquisition. However, it is possible to measure self-diffusion in systems with one or a few diffusing components in a steady magnetic field-gradient that is present during acquisition of the spin-echo, without Fourier transformation, provided that the components have sufficiently different self-diffusion coefficients. Steady gradient self-diffusion studies of polymer systems have been reviewed by von Meerwall [10].

The pulsed gradient experiment in its most basic form is also based on the simple Carr–Purcell spin-echo experiment. A magnetic field-gradient of magnitude G (units: T m^{-1}) is generated by special coils present in a probe designed for self-diffusion measurements. It is turned on for a short time δ between the 90° and the 180° RF pulses. After the 180° RF pulse and a total time elapse counting from the beginning of the first gradient pulse of duration Δ (usually of the order 100 ms for liquid systems using protons (see below)), a new pulsed field-gradient identical in magnitude and duration to the first one, is applied to the system. The key feature is that it is turned off before the echo is formed, and the acquisition of the second half of the echo is performed without any magnetic field-gradient present. This basic experiment is illustrated in Fig. 1. Since the magnetic field during the influence of the magnetic field-gradient varies linearly along the sample (typically in the z-direction) according to $B = B_0 + G \times z$, where B_0 is the magnetic field without any gradient, every spin in the sample acquires a phase angle in the rotating frame of reference of $\theta = G \times z \times \delta$. The 180° RF pulse inverts the precession direction of the individual spins, and if these do not move in space during the time interval Δ, the spins will refocus at the time 2τ and loose intensity only according to unavoidable spin–spin (T_2) relaxation effects (see above). However, if the spins move to another point in space because of self-diffusion during Δ, this will produce a loss of echo intensity because the spins will not refocus completely at 2τ. This loss may be calculated theoretically by taking into account that molecules will be displaced a root-mean-square distance Z (in z-direction) in a time interval t due to self-diffusion according to $Z^2 = 2 \times D \times t$, where D is the self-diffusion coefficient of the molecule containing the nucleus. The theoretical result for the echo intensity I is given by Eq. (8). See also Section 3 and Ref. [7]. In Eqs. (7) and (8), γ is the magnetogyric ratio of the nucleus under study, which in most cases is the proton, ^{1}H. This nucleus has the next highest γ value for all nuclei (only the radioactive tritium nucleus, ^{3}H, has a higher γ value), and hence has a high sensitivity for diffusion (cf. Eqs. (7) and (8)). It is also present in most polymers, with usually relatively long spin relaxation times. See more on this point, below. It deserves to be mentioned that the corresponding result for a steady gradient is contained in Eqs. (7) and (8) by letting $\delta = \Delta = 2\tau$.

Pulsed field-gradients may in modern commercial NMR instruments be produced with strengths of the order of 10 T m^{-1} (i.e. 1000 Gauss cm^{-1}), or even larger. This makes it possible to determine diffusion coefficients in the range 10^{-9}–10^{-14} m^2 s^{-1}, covering the range of self-diffusion coefficients that may ordinarily be expected in liquid-like systems.

By keeping the time interval τ constant in the experiments the effect of relaxation on reducing the spin-echo amplitude is constant (see Section 3). Therefore, the attenuation of this intensity as a function of changing either G or δ, comes about only as a result of self-diffusion of the molecules that the spins belong to. In the case of restricted self-diffusion, it may be shown that the equivalence between changing either G or δ no longer holds, and G is the proper experimental parameter to vary [11]. More on the topic of restricted self-diffusion is provided below. It is important that temperature gradients are absent in the sample, since these may cause convection in liquid samples. Yoshida and coworkers have recently constructed a probe for high-temperature pulsed field-gradient NMR work that eliminates problems with convection even in such low-viscosity samples as pure water [12].

One problem using ordinary Hahn-echoes in pulsed field-gradient work, i.e. echoes formed using the conventional 90°–τ–180° RF pulse sequence shown above, is that protons in polymers often have short spin–spin relaxation times, T_2, but considerably longer spin–lattice, T_1, relaxation times. This fact warrants the use of so-called stimulated echoes instead [13]. Here, a series of three 90° RF pulses are used, and the main point is that the spins relax along the z-direction, that is, relax according to the spin–lattice relaxation time T_1 during the diffusion observation time, Δ. There is a loss, however, in intensity of the stimulated echo as compared to the ordinary Hahn echo formed after the same time interval 2τ. Therefore, T_1 must in general be of the order of three times longer than T_2 to obtain any echo signal intensity recovery using this RF pulse sequence. However, this criterion is often met with polymers in the liquid state, as found in polymer solutions, melts or gels. The first magnetic field-gradient pulse must occur in the time interval between the first and the second 90° RF pulses, as illustrated in Fig. 1b. This means that the duration of the field-gradient pulses, δ, has to be very short, in practice of the order of 1–3 ms, since the spins relax according to their T_2 during this period. Therefore, very intense magnetic field-gradient pulses are generally needed for such studies of the self-diffusion of polymer chains.

Eddy-currents are formed in the probe chassis due to the very quick changes of magnetic field during start and stop of (nearly) rectangular pulses, and the eddy-currents produce distortions in the echoes unless this effect is counteracted. Detailed discussions on how to improve the situation with regard to eddy-currents can be found in the monographs of Callaghan [11] and Price [24].

As noted above, during the time interval Δ the spins (molecules) diffuse a mean-squared displacement (MSD) Z^2 in the z-direction (i.e. the direction of the magnetic field-gradient). As an example, for a molecule with self-diffusion coefficient of 10^{-11} m^2 s^{-1} observed for Δ = 0.1 s (100 ms), $(Z^2)^{1/2} \cong 10^{-6}$ m. The molecule will thus diffuse a micrometer in this direction during the observation time in a PGSE experiment. This distance is much larger than the dimension of most molecules and usually larger than most supramolecular structures met in chemical systems. The "nano-world", or the domain of classical colloidal systems, is 10^{-9}–10^{-6} m approximately. This means that measured self-diffusion coefficients of molecules in organized systems may contain information on structures on those length scales in such systems [14].

2. Scope of the present review

In this review we present a summary of field-gradient NMR work on polymer systems, with some emphasis on systems that can be grouped under the topic "soft polymer systems" and on work that has been published in the last two decades. We also include studies on polymer solutions, since a lot of theoretical models for polymer and solvent dynamics have been tested on such systems, that are in many respects simpler than more organized systems. We do not claim to provide a complete up-to-date review of this research area by covering all literature over the last two decades or so. However, we hope that the cited works, along with some comments by us will give an idea of the kind of detailed information on the diffusional dynamics that is presently obtainable in these types of system. Many review papers that partly cover topics included here exist, and should be consulted to get a wider scope of the research field.

2.1. Previous reviews on field-gradient NMR in polymer and colloidal systems

Stilbs has reviewed the state of the art of the pulsed field-gradient NMR technique and its applications in physico-chemical systems until 1987 [15], while Nose has reviewed pulsed field-gradient NMR studies of chain molecules in polymer matrices, until 1993 [16]. Söderman and Stilbs have reviewed NMR studies of surfactant systems, including pulsed field-gradient self-diffusion studies of adsorption of surfactants to polymers [14]. Lindblom and Orädd have reviewed NMR translational diffusion studies of lyotropic liquid crystals and lipid membranes [17]. Price has reviewed gradient NMR studies more generally, including a short paragraph on studies of polymers and macromolecules [18], while Yasunaga [19] has reviewed NMR spectroscopy methods, including pulsed field-gradient studies, of polymer gels. Of more recent reviews close to the topic of the present one may mention two by Masaro and Zhu: a short one in 1998 [20], discussing topics like restricted diffusion and anomalous diffusion, and a longer one in 1999 discussing physical models of diffusion in polymer solutions, gels and solids [21], and one by Yamane et al. [22] from 2006, describing *inter alia* works on diffusion of solvents and amino acids in polystyrene gels. Stallmach and Galvosas [23] have reviewed the state of developments of the various spin-echo NMR techniques, up to 2007.

Of general textbooks in this field the one by Callaghan [11], and a very recent one by Price, covering the theory of NMR diffusion measurements and the experimental details of present-day PGSE work [24], must be mentioned.

3. Pulsed field-gradient experiments

3.1. Background

The PGSE experiment is commonly based on either a two-pulse spin-echo or a three-pulse stimulated echo RF pulse sequence as shown in Fig. 1. The NMR signal is encoded for molecular displacements using magnetic field-gradient pulses with duration δ and amplitude G inserted into the defocusing and refocusing part of the RF pulse sequence. The time between the leading edges of the gradient pulses is denoted Δ. For an isotropic and homogeneous system, the evolution of the transverse magnetization density $m(\mathbf{r},t)$ is given by the Bloch–Torrey equation

$$\frac{\partial m(\mathbf{r},t)}{\partial t} = -i\gamma \mathrm{G}\cdot\mathbf{r}f(t)m(\mathbf{r},t) + D\nabla^2 m(\mathbf{r},t), \tag{1}$$

where γ is the magnetogyric ratio, $f(t)$ is the effective time modulation of the gradient, and D is the self-diffusion coefficient. The measured NMR signal is proportional to

$$M(t) = \int m(\mathbf{r},t)\mathrm{d}\mathbf{r}. \tag{2}$$

Using the condition of complete refocusing

$$\int_0^t f(t')\mathrm{d}t' = 0, \tag{3}$$

the solution of Eq. (1) yields

$$M(t) = M(0)e^{-bD}, \tag{4}$$

where

$$b = \gamma^2 G^2 \int_0^t \left(\int_0^{t'} f(t'')\mathrm{d}t''\right)^2 \mathrm{d}t'. \tag{5}$$

Inclusion of a relaxation term $-m(\mathbf{r},t)/T_2$ in Eq. (1) gives rise to a factor $\exp(-t/T_2)$ in Eq. (4), where T_2 is the transverse relaxation time; b is the field-gradient parameter.

The effective gradient modulation shown in Fig. 1(c) can be expressed as

$$f(t) = -H(t) + H(t-\delta) + H(t-\Delta) - H(t-\Delta-\delta), \tag{6}$$

where $H(t)$ is the Heaviside step function. Insertion of Eq. (6) into Eq. (5) yields

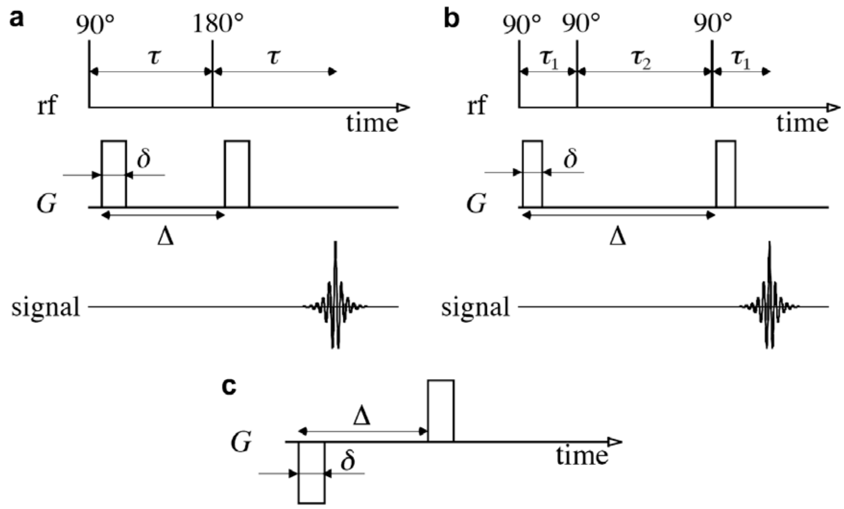

Fig. 1. Schematics of the pulsed gradient spin-echo (PGSE) experiment. (a) Spin-echo version with one 90° and one 180° RF pulse. (b) Stimulated echo version with three 90° pulses. (c) Effective gradient modulation $f(t)$ taking into account the phase-inverting effect of the RF pulses. The gradient pulses have the duration δ and the amplitude G. The time Δ is that between the onset of the gradient pulses.

$$b = \gamma^2 G^2 \delta^2 (\Delta - \delta/3). \tag{7}$$

The factor $(\Delta - \delta/3)$ can be interpreted as the effective time-scale, the diffusion time t_d, over which molecular displacements are monitored.

The PGSE experiment is performed by recording the NMR signal I as a function of b by varying G at constant values of all timing variables, thereby keeping the relaxation factors constant. The experiment is evaluated by fitting

$$I(b) = I_0 e^{-bD} \tag{8}$$

to the experimental data using D and the $G = 0$ intensity I_0 as adjustable parameters. A plot of log I vs. b is referred to as a Stejskal–Tanner plot, which yields a straight line for diffusion of a monodisperse species in a homogeneous system.

Eq. (8) is exact for the case of Gaussian diffusion. For other situations (such as in restricted diffusion) approximations are invoked. Within the so-called short gradient pulse limit, in which the gradient pulses are assumed infinitely short, a master equation for the decay of the echo intensity, I, can be written as

$$I = \int P_S(Z|\Delta) exp(i2\pi qZ)\mathrm{d}Z, \tag{9}$$

where $q = \frac{\gamma G\delta}{2\pi}$ and $P_S(Z|\Delta)$ is the average propagator describing the probability that a molecule has moved a distance of Z in time Δ irrespective of starting position. An interesting and useful property of Eq. (9) is obtained by expanding it for low values of q as

$$I = 1 - 2\pi^2 q^2 \langle Z^2 \rangle + O(q^4). \tag{10}$$

Eq. (10) implies that within the short gradient pulse limit, the MSD can be obtained from the initial slope of the echo intensities when plotted vs. q^2, irrespective of the nature of the diffusion process.

3.2. NMR PGSE applied to polymer solutions

In this section we consider the particular features that have to be considered when performing NMR diffusometry experiments on polymer solutions. The main problem arises because polymer systems are almost invariably polydisperse in nature, i.e. they contain a distribution of molecular sizes (see Section 1). We shall start by assuming that each species in the distribution has a diffusion coefficient equal to that it would have if it were alone in the solution under otherwise identical conditions. This implies dilute conditions, with concentrations below the overlap concentration, c^*. Later we will discuss deviations from this condition.

For a polydisperse system Eq. (8) for the echo attenuation is no longer valid and we must write [25,26]:

$$I = \frac{\sum p_i M_i \exp(-bD_i)}{\sum p_i M_i}, \tag{11}$$

where p_i is the weight of each species of molar weight M_i. We have assumed that there is no dependence of the value of T_2 on the polymer molecular weight. Expressing this in integral form we get:

$$I = \frac{\int MP(M)\exp(-bD)\mathrm{d}M}{\int MP(M)\mathrm{d}M}. \tag{12}$$

Here $P(M)$ is the molecular weight distribution (MWD). Although the actual form of the MWD function in general is not known, a common distribution function used to describe polydispersity is the log-normal distribution:

$$P(M) = \frac{1}{M\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(M) - \ln(M_0))^2}{2\sigma^2}\right), \tag{13}$$

where M_0 is the median molecular weight and σ is a measure of the width of the MWD.

The number, M_n, and weight, M_w, averages of Eq. (13) are $M_n = M_0 \exp\left(\frac{\sigma^2}{2}\right)$ and $M_w = M_0 \exp(\frac{3\sigma^2}{2})$, respectively. The polydispersity index is therefore:

$$\frac{M_w}{M_n} = \frac{\int M^2 P(M)\mathrm{d}M}{(\int MP(M)\mathrm{d}M)^2} = \exp(\sigma^2). \tag{14}$$

A scaling relation is often used to describe the dependence of the diffusion coefficient on the molecular weight:

$$D = KM^{-\alpha}, \tag{15}$$

where K and α are system dependent positive parameters, which have to be determined in separate experiments using near monodisperse reference samples (for a system below the overlap concentration, α is typically around 0.5–0.6 in a good solvent [25]).

By combining Eqs. 12, 13 and 15, it is possible to investigate the influence of the degree of polydispersity on the echo intensities. This is best done by introducing $L = \ln(M)$. We then obtain:

$$I = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(L) \exp\left(-\frac{(L-L_0)^2}{2\sigma^2}\right) \exp(-bK \times \exp(-\alpha L))\mathrm{d}L. \quad (16)$$

A numeric solution of the integral in Eq. (16) yields the results displayed in Fig. 2.

For a sample containing one species with a unique size, the Stejskal–Tanner representation of the data, i.e. a plot of log (I) vs. b, in Fig. 2 yields straight lines. Clearly, for a polydispersity index of 1.1, the deviation from linearity is slight, but increases for higher indexes. The degree of deviation also depends on the actual (mean) molecular weight. Svensson et al. have suggested that by plotting the intensity I vs. $b\langle D\rangle$ (where $\langle D\rangle$ is the average diffusion coefficient, see below) the effects of the polydispersity can be more clearly identified [27]. The reason for this is that the initial slope of the echo-decay can be shown to yield $\langle D\rangle$ (see below) and thus with $b\langle D\rangle$ on the abscissa the echo-decays all have a common initial decay. This is illustrated in Fig. 3.

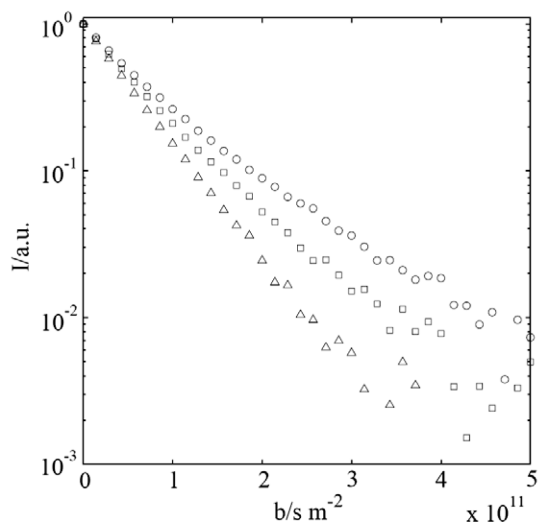

Fig. 2. A numeric solution to Eq. (16). The (normalized) echo intensities, I, are plotted vs. the field-gradient parameter b. Parameters used are: M_0 = 100 kDa, α = 0.525 and $K = 8.4 \times 10^{-9}$ m^2 s^{-1}. The polydispersity index is 1.1 (△), 1.5 (□) and 2 (○). A signal to noise ratio of 500:1 has been added to the data by generating normally distributed random numbers.

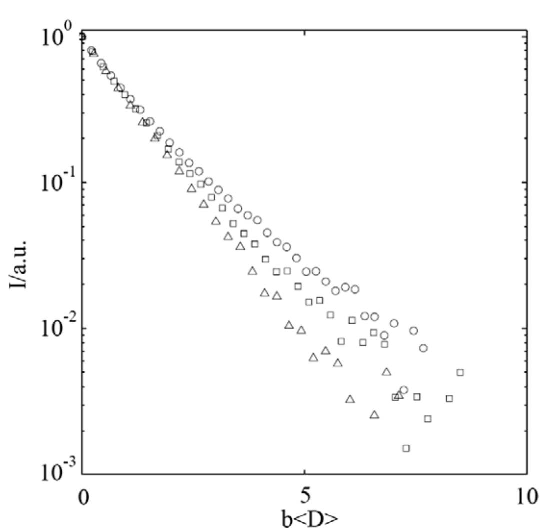

Fig. 3. The echo-decays in Fig. 2 plotted vs. $b\langle D\rangle$ to emphasize the effects due to polydispersity. Symbols as defined in Fig. 2.

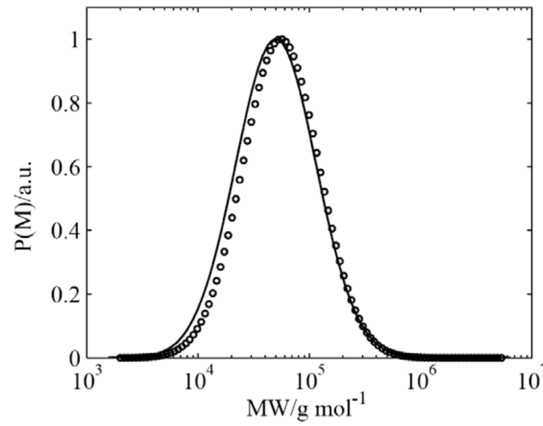

Fig. 4. The results of fitting Eq.(16) to one of the echo-decays in Fig. 2. The solid line is the input distribution (with M_0 = 100 kDa and polydispersity index of 2) while (○) are the predictions of the fit. The parameters of the predicted distribution are: M_0 = 95.5 ± 2.4 kDa and the value of the polydispersity index is 2.09 ± 0.04.

Often one is interested in the MWD. Given independent values of the scaling parameters K, α, and assuming that the MWD can be described by a log-normal function, the parameters of the distribution can be obtained from a non-linear least square fit to the echo-decay. This is shown in Fig. 4, which demonstrates that it is indeed possible to obtain a MWD from an echo-decay with reasonable accuracy, provided that the scaling parameters are known.

Sometimes it is convenient to express I in terms of a (normalized) distribution of diffusion coefficients $P(D)$:

$$I = \int P(D)\exp(-bD)\mathrm{d}D, \quad (17)$$

where a log-normal distribution of diffusion coefficients (DDC) is often used:

$$P(D) = \frac{1}{D\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(D)-\ln(D_0))^2}{2\sigma^2}\right). \quad (18)$$

The moments of the DDC can be obtained by expanding Eq. (17) at low values of b:

$$I = \int P(D)\left(1 - kD + \frac{b^2}{2}D^2 - \frac{b^3}{6}D^3 + \dots\right)\mathrm{d}D$$
$$= 1 - b\langle D\rangle + \frac{b^2}{2}\langle D^2\rangle - \frac{b^3}{6}\langle D^3\rangle \quad (19)$$

We note that the initial slope of the echo-decay yields the average self-diffusion coefficient $\langle D\rangle$. Using the DDC of Eq. (18) the averages of Eq. (19) can be expressed as $\langle D\rangle = D_0\exp(\frac{\sigma^2}{2})$, $\langle D^2\rangle = D_0^2\exp(2\sigma^2)$ and $\langle D^3\rangle = D_0^3\exp(\frac{9\sigma^2}{2})$, respectively.

$P(D)$ can also be obtained from an inverse Laplace transformation (ILT) of Eq. (17), without the need to assume a functional form of the DDC. While Laplace transformation is well-defined and a relatively straightforward mathematical procedure, the inverse process is mathematically ill-conditioned and care must be exercised in using this approach. This is also discussed in Refs. [28,29].

In Fig. 5 we show the Stejskal–Tanner representation of a PGSE experiment of a polymer with a log-normal DDC. Also shown are the results of a non-linear least square fit of the data to Eq. (17) and (18), and the results of a non-linear Laplace inversion of the echo-decay, using an algorithm based on the work presented in Refs. [30,31]. While the non-linear least square fit reproduces the input DDC, the result of the ILT is somewhat poorer.

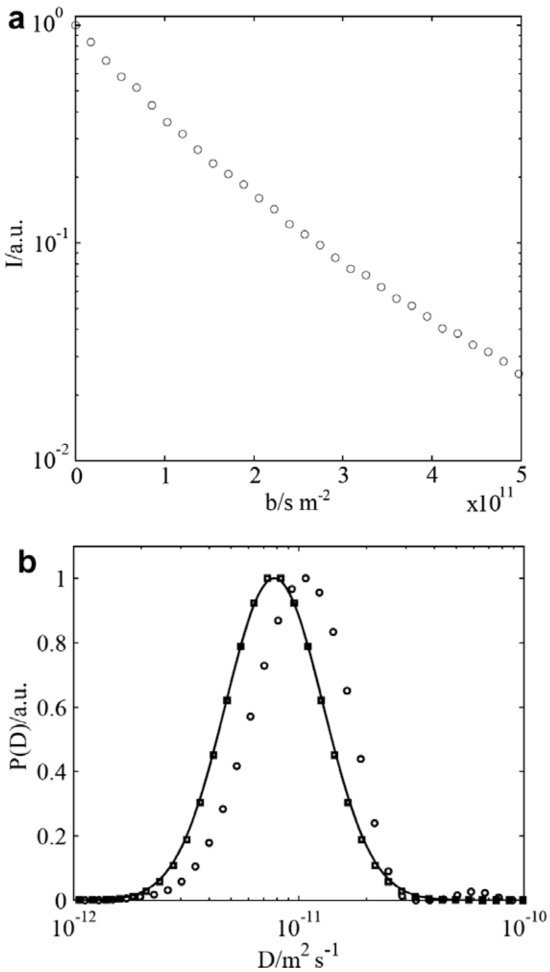

Fig. 5. (a) A calculated Stejskal–Tanner plot of echo intensity I versus the factor b in Eq. (7) for a polymer described by a log-normal distribution in diffusion coefficients with $D_0 = 1 \times 10^{-11}$ m^2 s^{-1} and $\sigma = 0.5$, obtained from Eqs. (17) and (18). (b) The solid line is the input distribution of self-diffusion coefficients (DDC). Also shown are results of fitting the echo-decay in Fig. 5(a) to a log-normal form of DDC in Eq. (7) (□). The obtained parameters are: $D_0 = 9.9 \times 10^{-12} \pm 0.07 \times 10^{-12}$ m^2 s^{-1} and $\sigma = 0.49 \pm 0.013$. Also shown in Fig. 5(b) are the results of a Laplace inversion of the echo-decay (○).

Finally we note that the MWD is related to the DDC through:

$$\frac{MP(M)\mathrm{d}M}{\int MP(M)\mathrm{d}M} = P(D)\mathrm{d}D, \tag{20a}$$

or, equivalently:

$$MP(M)\mathrm{d}M = P(D)\mathrm{d}D \int MP(M)\mathrm{d}M. \tag{20b}$$

From the scaling relation in Eq. (15) one obtains:

$$M^2P(M) = \mathrm{abs}(-\alpha)DP(D) \int MP(M)\mathrm{d}M. \tag{21}$$

It is thus possible to calculate the physically more interesting property MWD from the experimentally obtained DDC.

Sometimes a stretched exponential (often referred to as a Kohlrausch–Williams–Watts (KWW) function) is used to describe a non-linear Stejskal–Tanner plot:

$$I = \exp\left(-(bD)^{\beta}\right). \tag{22}$$

This approach has the advantage that it is straightforward to fit Eq. (22) to an experimental echo-decay and to obtain the stretching exponent β and a value of D (note that D is an effective diffusion coefficient, and should be considered only as a fitting parameter). In fact, it can be shown that:

$$\left\langle \frac{1}{D} \right\rangle = \frac{1}{D\beta}\Gamma\left(\frac{1}{\beta}\right), \tag{23}$$

where Γ is the gamma-function. Note that in general $\langle D\rangle \neq \langle \frac{1}{D}\rangle^{-1}$. For instance, for a log-normal distribution

$$\left\langle \frac{1}{D} \right\rangle^{-1} = D_0 \exp\left(-\frac{\sigma^2}{2}\right), \tag{24}$$

while, as noted above, $\langle D\rangle = D_0 \exp\left(\frac{\sigma^2}{2}\right)$. We also note that $\langle \frac{1}{D}\rangle$ can be obtained from a (numerical) integration of the echo-decay since:

$$\int_0^\infty I\mathrm{d}b = \int_0^\infty \int_0^\infty P(D)\exp(-bD)\mathrm{d}D\mathrm{d}b = \int_0^\infty P(D)\left(\frac{1}{D}\right)\mathrm{d}D = \left\langle \frac{1}{D} \right\rangle. \tag{25}$$

In the above outline we have treated the ensemble of polymer species as a sum of independent particles, with a diffusion coefficient equal to that of the isolated species alone. This is an approximation that becomes better the more dilute the system is. Consider the velocity field caused by the (slow) motion of a sphere of radius R moving though a solvent with velocity v. In a polar coordinate system (r, θ) centered on the sphere, the radial and angular components, respectively, of the velocity field are given by [25]:

$$v_r = v\ \cos(\theta)\left(1 - \frac{3}{2}\left(\frac{R}{r}\right) + \frac{1}{2}\left(\frac{R}{r}\right)^3\right) \tag{26}$$

$$v_\theta = v\ \sin(\theta)\left(-1 + \frac{3}{4}\left(\frac{R}{r}\right) + \frac{1}{4}\left(\frac{R}{r}\right)^3\right). \tag{27}$$

Although the detailed nature of these equations need not concern us here, it is clear that the velocity field decays as $\frac{1}{r}$ at long distances. As a consequence, a particle present in the velocity field of another particle will experience a force even at rather long distances from the "center" particle; this effect is usually termed the hydrodynamic interaction. Another effect is that particles surrounding the "center" particle will obstruct the propagation of the velocity field. This effect is referred to as hydrodynamic screening.

We end this section with a few remarks on the diffusion behaviour in the semidilute and concentrated regime where the polymers overlap and become entangled. A theory for self-diffusion in this concentration regime was suggested by De Gennes [32] and later further developed by Doi and Edwards [6]. The theory is based on a mechanism of reptation, where the polymer moves through a series of worm-like motions in a volume of obstructions set-up by the neighboring polymers. In the long-time limit when the diffusion is Gaussian the theory predicts that D scales with the molecular weight as M^{-2}. The reptation theory predicts four different regimes with different diffusion mechanisms as the measuring time over which the diffusion is followed is varied. Each regime predicts a certain value for the scaling exponent in the expression relating the mean-square dependence to the diffusion time:

$$\langle Z^2\rangle = t_\mathrm{d}^n, \tag{28}$$

a property that can be determined by means of PFG NMR (see above). The experimental evidence in favor of the reptation mechanism is quite convincing in the case of polymer melts, and Callaghan and Coy were able to show by using PGSE NMR that it also holds for polymers in the semidilute regime [33].

3.3. Instrumental development and hardware

During the last decade there has been an impressive development of the instrumentation for performing PGSE experiments, not the least assisted by the developments in the general field of NMR spectroscopy and in the related field of magnetic resonance imaging (MRI). It is not our intention to review these developments or to discuss technical aspects of NMR PGSE here. Good up-to-date accounts can be found in Refs. [24,34].

4. Applications

4.1. Polymer solutions

With "polymer solutions" we refer to two-component systems with polymeric molecules dissolved in a (usually) low molecular weight solvent. Depending on the nature of the polymer, i.e. the chemical composition of its repeating units or monomers, whether it is a homopolymer with only one type of repeating unit, or it is a copolymer with more than one (usually two) different units with or without block structure etc., the choice of solvent is critical. Most PGSE NMR investigations in such systems are performed on polymers in good solvents, i.e. where the second virial coefficient $A_2 > 0$, making the solutions thermodynamically stable (see Section 1). However, for block-type polymers, usually a certain solvent is a good one for only one of the blocks, and a bad solvent for the other type of blocks. In such a case, one often sees polymer aggregation behaviour, leading to associative (intra- or intermolecular) structures in the solutions. With development of NMR equipment capable of generating very high magnetic field-gradients G (see Eqs. (7) and (8)), it is possible to measure very slow self-diffusion, which is often the case for high molecular weight polymers in a good solvent. For spherical particles with a hydrodynamic radius r_H at infinite dilution, the Stokes–Einstein equation predicts a self-diffusion coefficient D equal to:

$$D = \frac{k_B T}{6\pi\eta r_H}; \tag{29}$$

k_B is the Boltzmann constant, T the absolute temperature and η is the viscosity of the medium (solvent).

4.1.1. Polymer shape and size

Polymers are usually not spherical at good solvent and temperature conditions. Often they can be described as either a prolate or oblate ellipsoid of revolution. Perrin [35] has derived expressions for the friction factor f in $D = k_B T/f$ as a generalization of Eq. (29) for the self-diffusion coefficients for particles (polymers) of such shape:

$$f = \frac{6\pi\eta a\left[\left(\frac{b}{a}\right)^2 - 1\right]^{1/2}}{\tan^{-1}\left[\left(\frac{b}{a}\right)^2 - 1\right]^{1/2}} \tag{30}$$

and

$$f = \frac{6\pi\eta b\left[1 - \left(\frac{a}{b}\right)^2\right]^{1/2}}{\ln\left\{\frac{1+\left[1-\left(\frac{a}{b}\right)^2\right]^{1/2}}{\frac{a}{b}}\right\}}. \tag{31}$$

Eq. (30) pertains to oblates and Eq. (31) to prolates; a and b denote the semi-minor and semi-major axes, respectively, of the ellipsoid.

In a number of papers Callaghan and coworkers have addressed the problem of polymer size and shape in various solvents by determining self-diffusion coefficients for the polymers and the solvent by pulsed field-gradient NMR [25,26,36,37]. The polymers used were polysaccharides like amylopectin [25], dextran, and glycogen [26], so-called β-limit dextrin [36], and kraft lignins [37]. Various solvents were used, such as dimethylsulfoxide (DMSO) which is a good solvent for the three first-mentioned polymers, and (heavy) water, which is a less good solvent. In the last quoted work [37], water and deuterochloroform were used as solvents. Also, a rather recent work has appeared where comparison is made between data generated by dynamic light scattering (DLS) and pulsed-field-gradient NMR on solutions of (cationized) amylopectin in water and showing how the response of the systems to shearing and heating is detected by these methods [38].

By a combination of self-diffusion coefficients of the solvent and of the solvated polymer molecules it is possible to obtain information about the shapes of the macromolecules or aggregates of the macromolecules in the solution. Depending on intramolecular interactions, primary and secondary structures, thermodynamic conditions (good/bad solvent), and temperature, the polymer chains may take on a variety of shapes like more or less flat sheets, random coils, and helices, that may be rod-like. In less good solvents the macromolecules may also form supramolecular aggregates with various shapes. The solvent molecules will be subject to an obstruction effect in their Brownian movement or self-diffusion because the macromolecules form an inaccessible volume fraction of the solution. This obstruction effect may be described by the following expression [25]:

$$D^{\text{solvent}} = D_0^{\text{solvent}}\left[1 - w\left\{(\alpha - 1)(\overline{V_p}d_0 + h) + h\right\}\right] \tag{32}$$

where D^{solvent} and D_0^{solvent} denote the measured solvent self-diffusion coefficient at polymer concentration w, in units of polymer weight fraction, and the corresponding self-diffusion coefficient in pure solvent at the same temperature, respectively. $\overline{V_p}$ is the apparent macromolecular specific volume, d_0 the density of the pure solvent, and h denotes the so-called solvation coefficient, defined as (mass solvent)/(mass solute). This last-mentioned parameter describes the solvation (hydration in aqueous solutions) of the polymer chains, in the so-called non-draining limit (Zimm model). All these quantities may be determined independently, and the only remaining unknown in Eq. (32) is the parameter α. It contains the information on the shape of the macromolecules or supramolecular aggregates that may be present in the solution. Analytical expressions for α in terms of the axial ratio $\rho = b/a$ for prolate or oblate shapes exist. These may be found in [25] or references cited therein. As pointed out in Ref. [25], the obstruction effect felt by the solvent molecules can not say anything about the actual size of the polymer molecules or aggregates. The solvent obstruction is a function only of the total obstructing volume and can not discriminate between the cases of few but large volume units or many but much smaller ones. By measuring also the polymer self-diffusion, the size (and shape) of these units may be determined. In an important work by Jönsson et al. [39] the theoretical dependence of the obstruction of small solvent molecules by larger colloidal particles of various shapes on the volume fraction of these obstructing units is given and displayed in Fig. 4 of that work. There is a large difference in obstruction between spherical or prolate formed aggregates on the one hand, and oblate formed particles on the other hand. Therefore, it is possible to distinguish between these cases by carefully studying the self-diffusion coefficients of the solvent as a function of the obstructing volume fraction. In a recent work on aqueous solutions of a polyoxyethylene trisiloxane surfactant this procedure has been used to determine the shape of the micelles formed [40]. It may also be mentioned that Johannesson and Halle have made very accurate simulations of obstruction phenomena in ordered (i.e. anisotropic) fluid systems [41].

4.1.2. Polydispersity effects

Callaghan and Pinder have addressed the problem of interpreting pulsed field-gradient spin-echo attenuation data for polymers that are not monodisperse [42]. Eq. (8) is valid for a component with a definite self-diffusion coefficient D for a molecule with hydrodynamic radius r_H, given by Eq. (29). The value of r_H scales with the number N of so-called statistical segments (of the order of a few monomers, and thereby proportional to the molecular weight) in a flexible chain according to $r_H \sim N^{\nu}$, where $\nu = 0.5$ in a theta solvent, and $\nu = 0.6$ in a good solvent (see also Eq. (15)). Polymer samples in general are characterized by a range of molecular weights that give rise to a corresponding distribution of molecular sizes when the polymer chains are dissolved in a solvent. This circumstance is reflected in a distribution of self-diffusion coefficients. The echo attenuation data from such systems therefore deviate from a single exponential, and a representative self-diffusion coefficient is therefore not directly accessible. There are several ways to get around this problem, but usually one has to resort to some model dependent analysis. Also, one should note that even when the spin-echoes are Fourier transformed, the problem still exists, because polymeric chains with the same chemical units or monomers, that differ only in the number of these monomers, give rise to identical NMR spectra. See also Section 3.

Callaghan and Pinder suggested Eq. (11) to describe the echo attenuation $I(b)$ for very dilute polymer solutions. Eq. (11) and its analogue, Eq. (17) are based on the assumption that the polymer chains with molecular weight M_i give rise to an NMR signal intensity that is directly proportional to M_i, i.e. proportional to the number of protons in the chains. Also, it is assumed that the spin–spin relaxation time T_2 is independent of this chain length. This may be a reasonable assumption for flexible chains, since the spin relaxation in that case is caused by local segmental motions within the molecules [43]. As noted above, in connection with Eq. (19), the initial part of the attenuation is determined by the weight average polymer self-diffusion coefficient for the solution.

The effect of polydispersity in polymer systems on the chain self-diffusion in solution has been the subject of several studies. Cosgrove and Griffiths have studied systematically the effect of mixing protonated polystyrene with deuterated polystyrene in carbon tetrachloride (good solvent) and deuterated cyclohexane (theta solvent). The polydispersity index was $M_w/M_n \leqslant 1.08$ for the polystyrene samples used in this study. By observing the proton signals, only the protonated chains will be "visible" during the PGSE measurements and the effect of polydispersity may be systematically studied by changing the properties such as molecular weight of the deuterated chains, and total polymer concentration [43]. A detailed study of friction coefficients operating in the dilute and semidilute concentration regions as a function of the molecular weights of the different polymer species, and solvent quality is presented. In part 2 of this study [44], only fully protonated polymers were used. In a rather qualitative way an attempt was made to determine a distribution p_i of diffusion coefficients by a somewhat arbitrary division of the echo attenuation into three domains, and by comparing the obtained three "diffusion coefficients" with values obtained from gel permeation chromatography determination of the real molecular weight distribution. The problem of determination of the distribution p_i of diffusion coefficients was clearly stated in this work [44]. However, several other papers have dealt with this problem, and different approaches have been made. One is the ILT method (see Section 3). This is utilized in the two-dimensional DOSY approach, suggested by Morris and Johnson [45,46]. Fourier transformed echoes from PGSE data give NMR signals from each component in the sample under investigation which can be treated by an ILT analysis that in principle may deconvolute the distribution p_i of self-diffusion coefficients for all the components. The data are finally displayed with the ordinary NMR spectrum of the sample along one axis, and with the corresponding distribution of diffusion coefficients along the other axis. A problem with this approach is the limited number of data points available for an ILT analysis from PGSE data. In the above cited work by Wassenius et al. this procedure has been used on PGSE data from amylopectin in aqueous solution [38].

Another approach is to model the distribution p_i with a log-normal distribution. In this case a distribution of diffusion coefficients is modeled according to Eq. (18) [47,48] where σ describes the width of the distribution of diffusion coefficients D around a central coefficient D_0. These two parameters describe the distribution, and may be determined by a fit of PGSE data to this distribution; see Section 3. In all attempts to unravel a distribution of diffusion coefficients it is important to limit the range of possible values for the self-diffusion coefficients. This range is determined by physical requirements imposed by the system under investigation, and by the experimental set-up, i.e. parameters like the range of field-gradients G used and the diffusion observation time t_d [46].

A third procedure that has been used to treat polydispersity effects in PGSE data are a generalization of Eq. (8) by the inclusion of a stretched exponential exponent β. See for example Ref. [49]. The phenomenological expression in Eq. (22) is used to fit PGSE data from systems where there is a deviation from single exponential decay. The point of departure is that the β parameter, with $0 < \beta \leqslant 1$, describes the deviation from a pure exponential decay, a procedure that works quite well. The problem is how to interprete the results of such a fit, which in addition to β gives an effective D (denoted D_{eff}). In analogy to interpretation of autocorrelation function data from DLS, the expression in Eq. (23) was suggested to be used in calculating a mean diffusion coefficient D_{mean} from β and D_{eff} [49]. This is a procedure that has been used previously in a number of studies, and a summary is given by Masaro and Zhu [20]. As discussed in Section 3, there is a problem connected with this procedure, since in general $\langle 1/D \rangle^{-1} \neq \langle D \rangle$ [47]. As a consequence, average self-diffusion coefficients obtained from Eq. (22) must be treated with care.

4.1.3. Associative structures

Associative structures are formed when block copolymers are dissolved in a solvent that is selective for one of the blocks. Such structures may also be formed in a mixture of two immiscible or partly miscible solvents which are selective for each of the blocks [50].

One type of block copolymers that has been studied by PGSE NMR is hydrophobically modified poly(ethylene oxide) (PEO) urethanes, where hydrocarbon chains are introduced at the ends of PEO blocks. This kind of block copolymer may in addition contain hydrophobic blocks between several PEO units [20,49,51,52]. Dissolving this kind of polymer in water, which is a good solvent for the PEO blocks at room temperature, and a bad solvent for the hydrocarbon blocks, results in association of the hydrocarbon blocks. The associations may be intramolecular and intermolecular, forming associative structures in the solutions. Such associations are not permanent, but are dynamic in nature. The viscosity of such solutions rises steeply with polymer concentration above a certain concentration close to the overlap concentration, c^*, due to these associative structures. Fig. 6 shows a schematic picture of the expected structures formed in aqueous solutions of diblock copolymers (with hydrocarbon ends only in one end of the PEO block), and triblock copolymers with hydrocarbon chains in both ends of the hydrophilic block. As shown in Fig. 7 the spin-echo attenuation of the polymers deviate from a single exponential decay, and the solid lines through the data represent fits to Eq. (22). The deviation from a single exponential decay, described by the parameter β, increases with polymer concentration, and is larger for the triblock system than for the corresponding diblock system.

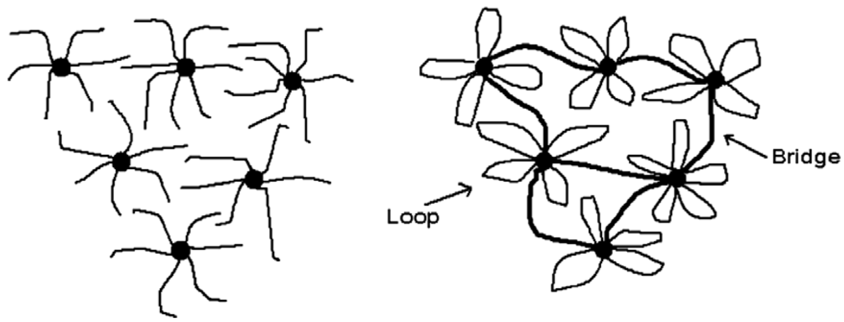

Fig. 6. Schematic representation of proposed structures of solutions of diblock (left) and triblock (right) polymers. Redrawn from Ref. [51] with permission.

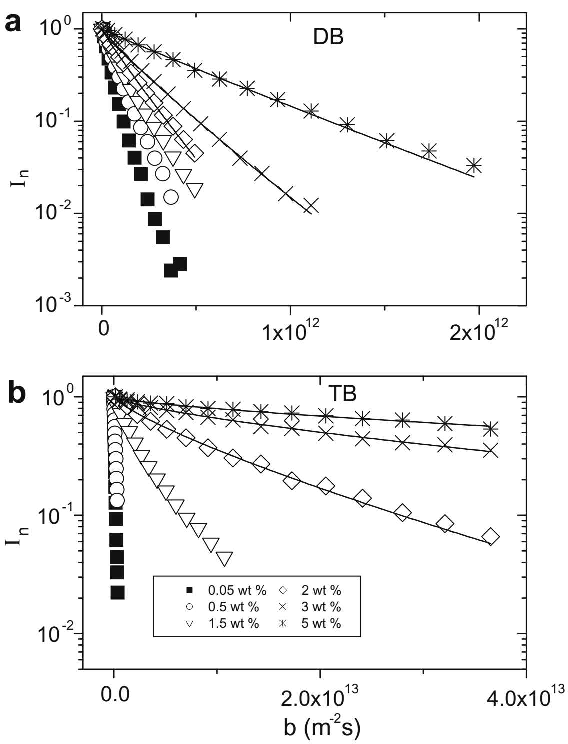

Fig. 7. Spin-echo attenuation in the systems and concentrations indicated. The curves are fits of Eq. (22) to the data. DB and TB denote diblock and triblock polymers, respectively. Redrawn from Ref. [51] with permission.

This non-exponential behaviour for the echo attenuation is ascribed to an expected polydispersity of the aggregates formed.

Triblock copolymers of the type poly(ethylene oxide)-poly(propylene oxide)-poly(ethylene oxide) (PEO-PPO-PEO), also called poloxamers or known under the trade name Pluronics, associate into micelles in aqueous solutions, because water is a better solvent for the PEO blocks than for the PPO blocks. However, this solvent effect is rather temperature dependent. Therefore, a minimum temperature, the so-called critical micelle temperature, cmt, is needed for micellization to occur. Also, the ability of the polymers to form micelles is limited by an upper temperature, where the PEO blocks are dehydrated (a cloud point phenomenon), and a phase separation eventually takes place. Gille et al. measured the temperature dependence of self-diffusion coefficients and have calculated hydrodynamic radii of micelles and monomers in dilute aqueous solutions of a PEO-PPO-PEO copolymer (i.e. Pluronic P85) [53]. Eq. (22) was used to fit the spin-echo attenuation data (not Fourier transformed), and some deviation from a single exponential decay was found. The parameter D in the stretched exponential function was used as a measure of the polymer self-diffusion coefficient in each sample. (See, however, the discussion above). The micellization process in the investigated solutions was demonstrated to take place over a rather wide temperature range, 10–15 °C. Also, the hydrodynamic radii of the micelles formed in H_2O were found to be ca. 25% larger than the corresponding radii of micelles formed in D_2O. This effect is attributed to stronger hydrogen bonds between ethylene oxide (EO) segments and D_2O than corresponding hydrogen bonds in H_2O. It was argued that this circumstance results in more compact EO chains in D_2O than in H_2O.

In a recent work the influence of polydispersity of the triblock copolymers (molecular weight and composition dispersity) on spin-echo raw data from PGSE NMR experiments was examined [54]. It was concluded that due to the presence of various types of polymers (with respect to polymer weight, ratio of block sizes, and the presence of diblocks) that are present in commercial samples of these chemicals, the degree of association varies among the different polymer species present. This circumstance gives rise to the observed departure from a single exponential decay in the spin-echo attenuation. Thus, the departure from a single exponential decay in the spin-echo attenuation may be regarded as an artifact and a reflection of the chemical heterogeneity of the samples.

Another class of polymers that form associative structures in aqueous solutions are chemically modified cellulose ethers, such as ethyl (hydroxyethyl) cellulose (EHEC). This class of polymers are water soluble, due to oxyethylene groups grafted on the cellulose backbone. Also, hydrophobic ethyl groups are sometimes grafted on the backbone, forming so-called "stickers". Depending on the degree of substitution of these two kinds of side-groups, the physical properties of the polymers may be varied to suit a specific requirement. In one study, a special hydrophobically modified EHEC was used (EHEC 100) [47]. Combined with ionic surfactants (see below), this polymer may form physical gels above a certain temperature. The manufacturing procedures produce polymer batches that are polydisperse in molecular weight, with typical molecular weights in the range from 100,000 to 200,000 Da. When dissolved in water, these molecules associate due to the hydrophobic side-groups, forming clusters of various sizes. These solutions therefore are quite heterogeneous also on the supramolecular level, as manifested in a rather complex dynamics that is seen in, for example, the pulsed field-gradient NMR results. See [47] and references cited therein. The deviation from a single exponential echo attenuation is large, even at rather low polymer concentration, and increases with increasing polymer concentration. Analysis of the data in terms of the log-normal distribution of diffusion coefficients (see above) shows that this distribution becomes much larger than can be accounted for by the inherent molecular weight distribution of the polymers themselves. It is concluded that some of the effect may be attributed to chains that do not associate. This is manifested in a decreasing width of the distribution, σ (see Eq.

(18)) with increasing temperature at constant polymer concentration, when the data are analyzed using that model. The spin-echo data are also analyzed using the stretched exponential approach, Eqs. (22) and (23), and compared with the results using the log-normal distribution method. It is demonstrated that the two approaches may give quite different measures of a mean or average diffusion coefficient D_m, in that Eq. (23) gives too high a weight to slowly diffusing species. The two approaches may lead to estimates of diffusion coefficients that differ by as much as a factor of 10. The complex dynamics in these kinds of polymer solutions is also found to be reflected in polymer proton spin–spin (T_2) relaxation data, in that the echo-decay as a function of time is non-exponential, and may be described by a sum of two stretched exponentials (stimulated echoes were used in corresponding PGSE measurements in these cases). We will return to other recent PGSE NMR studies of solutions and gels of aqueous EHEC systems below.

Konak et al. have studied the dynamics in solutions of a triblock copolymer polystyrene-*b*-poly(ethylene,butylene)-*b*-polystyrene (PS-PEB-PS) in the solvent *n*-heptane, which is a selective solvent for the middle block [55]. Depending on concentration "flower-like" micelles are formed, and also clusters of micelles of various sizes formed by bridging polymer chains between the micelles seem to be present in the dilute concentration range. This is manifested in DLS data that gradually becomes more complex with increasing polymer concentration. For very dilute solutions the autocorrelation function consists of a single diffusional mode ascribed to the existence of micelles. At higher concentrations within the dilute region, a second mode enters (also diffusive), that is ascribed to the existence of clusters of micelles, and is described by a stretched exponential function. In the semidilute region, three different modes in the autocorrelation function are revealed, with a nondiffusive mode in between the two outer modes that are still diffusive in nature. PGSE NMR echo attenuation data (the echoes are not Fourier transformed) in general show deviation from a single exponential decay. They are described using stretched exponentials, and the data are analyzed according to Eqs. (22) and (23).

Fleischer et al. have studied the dynamics of a polystyrene-*b*-polybutadiene-*b*-polystyrene triblock copolymer dissolved in *n*-heptane, which is a selective solvent for the middle block [56]. The association of the polymer molecules into micelle-like structures as a function of concentration, in a temperature range from 20 to 35 °C, is clearly manifested in the spin-echo data. These data show in general quite large deviations from a single exponential decay, due to the associative structures formed, and were subject to both a stretched exponential (see Eqs. (22) and (23)) analysis, and a fit to a log-normal distribution of diffusion coefficients. Some evidence of anomalous diffusion was found in the PGSE data, in that the apparent self-diffusion coefficients obtained are dependent on the diffusion observation time t_d (see Section 3): $D \propto t_d^{k-1}$, with $k < 1$. The effect was qualitatively ascribed to possible restricted diffusion and exchange between aggregated and nonaggregated states for the chains.

One more PGSE study of a block copolymer in a selective solvent that forms micelle structures should be mentioned. Fleischer et al. studied solutions of polystyrene-*b*-poly(ethylene-*co*-propylene) diblock copolymers in *n*-decane, that is selective for the aliphatic block [57]. It is concluded that micelles of these block copolymer chains are formed, with the polystyrene (PS) block forming a core and the aliphatic block forming the corona. At room temperature the PS blocks are in a glassy state. Therefore, heating the solutions to temperatures above the glass-transition temperature T_g for PS and cooling to room temperature, leads to a decrease of the core of the micelles, as evidenced by increased diffusivity for the micelles, measured by PGSE NMR (without Fourier transformation of the echoes). The PGSE attenuation data are found to be non-exponential due to a polydispersity of micellar size, and are analyzed using Eqs. (22) and (23). An important conclusion in that work is that the self-diffusion of the micelles, as a function of polymer concentration, may be described by a particular model for the self-diffusion of multi-arm star polymers, where the number of arms may be identified with the aggregation number of the micelles (see the cited work for details). Also, evidence for restricted or anomalous micellar self-diffusion (see below) at rather high concentration, close to a colloidal glass transition, was reported in that work.

Vanhoorne et al. have presented a study of solutions of ω-metal sulfonato PS in toluene at an elevated temperature of 56 °C [58]. It is concluded that micelles are formed in these solutions, with the ion-pairs forming a spherical core, and the PS forming a corona. The dynamics in these systems were studied by a combination of spin relaxation measurements, probing local dynamics for the metal ions in the nanosecond time-scale, and PGSE measurements. The PGSE measurements (with no Fourier transformation of the echoes) yielded self-diffusion coefficients based on single exponential echo attenuation data, and no deviation from this behaviour was reported in that work. An interesting effect reported is the dependence of the nature of the metal ion on the aggregation number and consequently the size of the micelles, in that Ba^{2+}-ions favor rather small micelles, whereas K^{+}-ions favor much larger micelles. Also, the influence of adding methanol to these solutions was investigated. Methanol will solvate the ions, but aggregates are still reported to exist even at a rather high degree of methanol addition.

A recent study of the dynamics and structure in solutions made by two partially miscible liquids, i.e. dimethylformamide (DMF) and cyclohexane (CH), and a diblock copolymer, i.e. polystyrene-*b*-poly(ethylene-*co*-propylene) (PS-PEP) has been presented [59a]. Dimethylformamide acts as a selective solvent for the PS block, while cyclohexane acts as a selective solvent for the PEP block. Depending on temperature, polymer concentration, composition of the polymer etc., such systems may give rise to a range of different supramolecular structures. Above the critical temperature of ca. 50 °C where the two liquids are completely miscible, the polymers are dissolved and form a polymer solution. Below this temperature, where the solvents will separate into two layers, it is concluded, based on DLS data, cryo transmission electron microscopy (cryo-TEM) measurements, small angle neutron scattering (SANS), and PGSE NMR on the solvent and polymer, that the system forms a microscopically segregated system with long-range order, with domains rich in CH and DMF, respectively, separated by interfaces consisting of diblock copolymer, see Ref. [59b]. The PGSE results are not displayed in a conventional way, and it is not stated in Ref. [59a] if the (stimulated) spin-echoes are Fourier transformed.

Aqueous solutions of methylprednisolone esters of hyaluronan at rather low concentrations have recently been examined by several techniques including PGSE NMR [60]. The echo attenuation data obtained for the solute molecules presented in this study seem to be single exponential as shown by the fits of the data to Eqs. (4) and (7). By comparing self-diffusion coefficients for molecules with various chemical composition in the dilute concentration region and calculating corresponding hydrodynamic radii according to the Stokes–Einstein equation (Eq. (29)), it has been possible to draw conclusions regarding internal hydrophobic associations between the molecules. The authors obtain a distribution of diffusion coefficients $P(D)$ (see Eq. (17)) using an ILT analysis (see above), and derive a distribution of corresponding molecular sizes over two orders of magnitude. It should be noted, however, that the ILT procedure can produce a distribution of diffusion coefficients even when the raw PGSE data can be described by a single exponential decay, see also Ref. [38]. Two other works that have used an ILT approach to study associative structures in aqueous solutions of polymers should be mentioned: one on hydrophobically modified poly(vinyl alcohol) [61], and the other on solutions of amino acid–based polymeric surfactants [62].

Gröger et al. have investigated co-micellization of the two block copolymers: polystyrene-*b*-polyisoprene (PS-PI) and polystyrene-*b*-polybutadiene-*b*-polystyrene (PS-PB-PS), in selective solvents [63]. The two (deuterated) solvents used were *n*-decane which is selective for the PI and PB blocks, and 1,4-dioxane, which is selective for the PS blocks. In 1,4-dioxane the diblock and the triblock copolymers form mixed aggregates, and the composition of these were determined from the self-diffusion coefficients of the components as a function of the over-all weight fraction of triblock copolymer (the echoes were not Fourier transformed, and the self-diffusion coefficients of the block copolymers were determined from bi- or tri-exponential echo attenuation data). A detailed picture of the composition of the mixed aggregates was demonstrated, and deviates from the corresponding picture generated from light scattering data (summarized in Fig. 3 in that work).

4.1.4. Polymer dynamics studies

Masaro and Zhu have summarized physical models that commonly are used in the analysis of diffusion data from polymer solutions [21]. They have categorized these models into three broad classes, *namely* diffusion models based on obstruction effects, hydrodynamic theories, and free volume models. In addition, they have also included some other types of models for polymer diffusion in their survey. In most of the cited works in this section, pulsed field-gradient NMR is used together with other methods to characterize the dynamics of polymer solutions. While DLS measures mutual diffusion, including internal dynamics, self-diffusion is a measure of the center-of-mass movement of the molecules in the solution, and important characteristics like hydrodynamic radii of the molecules (or molecular aggregates) may be determined using the measured self-diffusion coefficients.

In a contribution from the early 1990s, Brown and Zhou [64] measured the self-diffusion of polyisobutylene in chloroform and the theta solvent benzene to obtain friction coefficients for the polymer chains. By varying the molecular weight of rather monodisperse polymer samples, it was found that the hydrodynamic radii (r_H) scales with molecular weight (M) as M^{ν}, where the exponent ν was found to be close to 0.5 in the theta solvent, and close to 0.6 in the good solvent. These values for the exponent are close to those expected from renormalization group (RG) theory (i.e. 0.5 and 0.588, respectively) [6].

Also, from this period, Amelar et al. [65] found, partly based on results from PGSE NMR that the bead-spring model (BSM) well describes the dynamics of very low molecular weight polystyrenes in the solvent Aroclor 1248, at low concentrations. This was considered to be surprising, since this rather crude model for the dynamics of a polymer chain on a local scale seems to be valid for polymer chains with molecular weight that corresponds to a single bead-spring unit [65].

Pan et al. have compared DLS data from solutions of symmetric polystyrene-polyisoprene diblock copolymers (molecular weights M_w in the range 1.6×10^4–1.7×10^5 Da) in the neutral good solvent toluene [66]. Depending on concentration, the DLS correlation function may contain up to three relaxation modes, predicted theoretically using models for polymer dynamics. There is one mode corresponding to cooperative diffusion. In dilute solutions, this is found to be superimposed on a so-called heterogeneity mode that is relaxed by translational diffusion (or self-diffusion). In semidilute and concentrated solutions, these two modes are resolved. PGSE NMR is used to determine self-diffusion coefficients of the polymers in these solutions, and the predicted, close correspondence between the heterogeneity mode measured by DLS, and the self-diffusion of the polymer molecules is revealed. Also, as a continuation of this work, Liu et al. compared DLS data from solutions of a symmetric polystyrene-polyisoprene diblock copolymer (molecular weight $M_w = 3.4 \times 10^5$ Da) in a mixed, neutral good solvent that may index-match the whole copolymer, or either blocks, thereby eliminating the mutual diffusion mode from autocorrelation data [67]. In solvent conditions corresponding to zero-average contrast (ZAC), it was found that the correlation function consists of two modes both in dilute and semidilute solutions. Self-diffusion coefficients were found to correspond to coefficients calculated from the slow diffusive mode of the DLS correlation function.

Bandis et al. have studied penetrant diffusion behaviour in toluene-polyisobutylene solutions [68]. A range of temperatures (from room temperature to 70 °C) and volume fractions of toluene were covered. The self-diffusion of toluene was found to be in excellent agreement with predictions of free volume models by Fujita [69], and Vrentas, Vrentas and Duda [70].

Anastasiadis et al. have measured self-diffusion of diblock copolymers in a neutral good solvent [71]. The polymers, dissolved in toluene, were poly(styrene-*b*-1,4-butadiene) and poly(styrene-*b*-methyl metacrylate) copolymers of various molecular weights (polydispersity indices are in the range 1.03–1.06) and compositions, at room temperature. Spin-echo attenuation data were generally found to be single exponential, but in cases of small deviations from such behaviour, the data were subject to a cumulant analysis (see Eq. (19)) to determine the self-diffusion coefficients, measured as a function of molecular weight, composition, and concentration. A major conclusion from the work is that the difference in entanglement characteristics (i.e. critical molecular weight, M_c, or critical concentration, ϕ_c, for entanglement) of the parent homopolymers is important for the self-diffusion of the diblock copolymer chains. A progressive transition from Rouse ($D_s \sim \phi^{-0.53} M^{-1}$) to entanglement regime (where the concentration ϕ (i.e. volume fraction) exponent in theoretical expressions for D_s is much larger than -0.53) is paralleled with the dependence of the polymer self-diffusion on the diblock composition.

In a work by Chrissopoulou et al. polymer dynamics in toluene semidilute solutions of linear tetrablock and inverse starblock copolymers composed of styrene and isoprene were investigated by a combination of DLS and PGSE NMR [72]. A close correspondence between the diffusion coefficients of the so-called polydispersity mode in the DLS correlation function and the self-diffusion coefficients of the polymers determined by PGSE NMR was revealed. Moreover, it was demonstrated that the self-diffusion data obtained from these solutions as a function of polymer concentration may be given a uniform description (i.e. forms a master curve) by plotting the quantity$[D_s(gN)^{\nu}N_e^{1-\nu}]$ versus $[\phi/\phi_e]$ (Rouse-type dynamics); g is a parameter dependent on the polymer architecture, N_e denotes the number of segments corresponding to the average entanglement molecular weight, N is the total degree of polymerization, ϕ_e is the characteristic concentration for entanglements, and ν is the Flory exponent.

Anastasiadis et al. investigated MSD of polymer chains determined by PGSE NMR in solutions in toluene of very high molecular weight diblock copolymers composed of styrene and isoprene [73]. The copolymers were made by anionic polymerization, a procedure that makes them almost monodisperse. Using strong magnetic field-gradients G corresponding to $5.7 \times 10^{-4} \leqslant q \leqslant 0.0124\ \mathrm{nm}^{-1}$ (see Eq. (9)), it was possible to study polymer mean-square displacement over distances comparable to characteristic dimensions of the polymer molecules. Solutions of diblock copolymers of the kind studied in this work may undergo a disorder-to-order transition, depending on composition of the copolymer and concentration. It was demonstrated that $\langle Z^2 \rangle$ for the polymers scales with time according to:$\langle Z^2 \rangle \propto t_d^{1/2}$ (a subdiffusive motion) for times shorter than a characteristic time t_1, which for entangled chains is the disentanglement time, as predicted by theory [32,33]. Close to a disorder-to-order transition, there is evidence for a time domain succeeding the $t_d^{1/2}$ behaviour, where the mean-square displacement is independent of time until a characteristic time t_2. This effect is attributed to a trapping of the chains. For times longer than t_2 (dependent on polymer characteristics), the MSD was found to be diffusive, with $\langle Z^2 \rangle \propto t_d$.

In three papers, solutions of dendrimers have been investigated using PGSE NMR. One is devoted to five generations of poly(propylene imine) dendrimers dissolved in methanol, covering the whole concentration range, for three temperatures in the range 5–45 °C [74]. Hydrodynamic radii were determined, and the self-diffusion coefficients as a function of concentration were found to fit approximately to the free volume model of Vrentas, Vrentas and Duda [70]. The second contribution deals with PGSE NMR in aqueous solutions of three of these (hydrophilically modified) dendrimers in the presence of poly(vinyl alcohol) [75]. The self-diffusion coefficients of the dendrimers examined as a function of poly(vinyl alcohol) concentration, were found to fit the model for penetrant (sometimes called diffusant) diffusion in a polymer matrix, developed by Petit et al. [76]. A closely related work on diffusion of hyperbranched polyglycidols in aqueous solutions of poly(vinyl alcohol), measured as a function of polymer concentration, follows the same model for penetrant diffusion [77].

In a third work on dendrimers by Krykin et al. self-diffusion of poly(carbosilane) dendrimers of 3rd, 4th and 5th generations in carbon tetrachloride solutions were determined by PGSE NMR at high polymer concentrations at a temperature of 20 °C [78]. Spin-echo attenuation data were reported to be single exponential, but a dependence of the apparent self-diffusion coefficients on the diffusion observation time t_d is reported at dendrimer concentrations above ca. 20 wt% for short observation times. The apparent self-diffusion coefficients D_s, that were found to scale with Δ according to $D_s \sim \Delta^{-1}$, were used to calculate lengths l of so-called "high mobility regions, estimated from the relation $l = (2D_s\Delta)^{1/2}$. For polymer concentrations around 90 wt% values for l were found to be in the range ca. 300–600 nm, decreasing from 3rd to 5th generation dendrimers. Further information on anomalous/restricted diffusion is discussed below.

Chekal and Torkelson have examined the influence of chain length on the concentration dependence of polymer and oligomer self-diffusion in solutions [79]. Almost monodisperse samples of poly- and oligo-styrenes (PS) were dissolved using styrene and toluene as solvents. Self-diffusion coefficients for PS and solvent were determined as a function of concentration using PGSE NMR. In addition, literature values for self-diffusion coefficients of PS (D_p) in solvents such as benzene, tetrahydrofuran, and carbon tetrachloride, together with corresponding self-diffusion coefficients of the solvents (D_s), were used to obtain a universal description of these self-diffusion coefficients, in the form of a partly phenomenological relationship:

$$D_p(c)/D_p(0) = [D_s(c)/D_s(0)]^{\lambda}, \tag{33}$$

where $D_p(c)$ and $D_p(0)$ denote the PS self-diffusion coefficient at a PS concentration c and at infinite dilution, respectively, with corresponding definition of the solvent self-diffusion coefficients. The parameter λ is found to be dependent on the PS chain length, increasing from 1 for chain length of one unit, to ca. 2.3 for chain length of 20 units. The λ parameter is approximately constant for PS samples of between 20 and 1000 units chain length, but increases fast for longer chains, most probably because of entanglement effects. The temperature was kept at 25 °C, much below the glass-transition temperature T_g for PS. The form of the universal description given above for these self-diffusion coefficients may be rationalized within the Vrentas, Vrentas and Duda free volume model [70]. It should also be stated that self-diffusion coefficients of poly(methyl metacrylate) (PMMA) and toluene in PMMA/toluene solutions could also be fitted to Eq. (33). Further details are provided in Ref. [79].

Arcos-Casarrubias and Olayo have presented a study of PS in solutions of styrene [80]. Self-diffusion coefficients of solvent, of the initiator 2,2′-azobisisobutyronitrile (AIBN), and of PS formed by free radical bulk polymerization, were determined at various stages of polymerization. In addition, self-diffusion coefficients of a highly monodisperse PS and of solvent (toluene) were determined. Free radical polymerization gives rise to highly polydisperse samples of polymers, but the problem connected with determination of self-diffusion coefficients in polydisperse samples (see Section 3) does not seem to have been addressed. The solvent and AIBN self-diffusion data were found to fit to the model developed by Petit et al. [76], at least for dilute polymer solutions. The polymer self-diffusion data were fitted to phenomenological models, with little degree of physical interpretation.

Kanematsu et al. have presented a study of the relation between mutual- and self-diffusion coefficients of a semiflexible polymer in solution [81]. Mutual diffusion coefficients of the cellulose derivative cellulose tris(phenyl carbamate) in tetrahydrofuran were determined by DLS, and compared to self-diffusion coefficients of the polymer, determined by PGSE NMR. A renormalization of the mutual coefficients D_m, which defines a new diffusion coefficient $\tilde{D}$ according to:

$$\tilde{D} = D_m/[(M/RT)(\partial\Pi/\partial c)(1 - \bar{v}c)], \tag{34}$$

were performed to eliminate the thermodynamic driving force and the solvent backflow contributions to mutual diffusion in binary systems. M is the polymer molecular weight, c is the polymer mass concentration, $\bar{v}$ is the partial specific volume of the polymer, RT is the molar gas constant multiplied by the absolute temperature, and Π is the osmotic pressure of the solution. This procedure makes the polymer self-diffusion coefficients quite similar to the $\tilde{D}$ coefficients over the concentration range and for the polymer molecular weights studied. Also, the polymer self-diffusion coefficients were fitted to a special model, called "fuzzy cylinder theory" model, which is based on a combination of free volume and hydrodynamic interaction arguments, and which is expected to work well for rather stiff polymers. It was found that one could fit the self-diffusion coefficients of low molecular weight fraction of the cellulose derivative with few Kuhn statistical segments to this model over a large polymer concentration interval. For higher molecular weight fractions, the concentration region where it is possible to fit the self-diffusion or renormalized mutual diffusion data to this model, becomes smaller.

Zhao et al. have measured the self-diffusion coefficients of poly((2,5-didodecyl-*p*-phenylene)ethynylene) (didodecyl-PPE) and poly(2,5-didodecyl-*p*-xylylene) (didodecyl-PPX) polymers in dilute (deutero)chloroform solutions [82]. A slight deviation from single exponential spin-echo attenuation was demonstrated and the data were generally fitted to a log-normal distribution of diffusion coefficients (see Eq. (18)). Infinite dilution self-diffusion coefficients were determined, and found to scale with the polymer molecular weight M according to theoretically expected exponent values of 0.50 (see Eq. (15)) for flexible random coil chains in the case of didodecyl-PPX, and of 0.71 in the case of didodecyl-PPE, indicative of semiflexible rod-like macromolecules. This conclusion was also found to be consistent with concentration scaling for the self-diffusion coefficients for these two types of polymers in chloroform solutions.

Shimada et al. have reported precise measurements of self-diffusion coefficients of poly(ethylene glycol) (PEG) in aqueous solution [83]. Monodisperse samples of PEG oligomers in the polymerization degree range 1–40 were prepared, and the oligomer self-diffusion coefficients were determined by PGSE NMR at 30 °C. As a function of molecular weight, the infinite dilution self-diffusion coefficients were found to follow the scaling relation: $D \propto M^{-0.43\pm0.01}$. The scaling exponent is somewhat less than that expected for Zimm-type dynamics (i.e. 0.50), and the authors point to the possibility that the deviation is due to strong hydrodynamic interactions caused by hydrogen bonding of water molecules (D_2O) to the PEG chains.

Zhang and Amsden have quite recently studied self-diffusion of *inter alia* proteins in semidilute (aqueous) alginate solutions [84]. The so-called Obstruction-Scaling, developed by Amsden, was found to describe the self-diffusion of proteins (at room temperature) as a function of alginate concentration, at least for large proteins where hydrodynamic interactions between the macromolecules may be neglected.

Cosgrove et al. have recently presented self-diffusion data of solvent [85] and of carboxylated acrylic random copolymers [86] composed of butyl methacrylate (BMA) and methacrylic acid (MAA) with a range of molar BMA/MAA ratios in isopropanol solutions. Solvent self-diffusion was found to be dependent on the composition of the copolymers, and could be fitted to Wang's model (see e.g. Ref. [21]) for obstruction of solvent self-diffusion. The amount of solvent bound was found to be dependent on the composition of the copolymers, increasing with mol% MAA in the copolymer [85]. The polymer self-diffusion data indicate anomalous diffusion (see also more on this topic, in Section 4.1.5), in that the apparent self-diffusion coefficients are found to be dependent on the observation time Δ (see Eq (7)). The mean-square displacement is found to scale with Δ according to: $\langle Z^2 \rangle \sim \Delta^k$, where k may, typically, take values such as 0.78 or 0.87 dependent on the polymer's composition, at a polymer concentration of ca. 32% (see also above). The anomalous diffusion is ascribed to a possibility for the polymer chains to either form and be a part of the network, or to be diffusing rather freely within this network. An exchange between these two states for the chains within the observation time may explain the observations [86].

4.1.5. Restricted or anomalous diffusion

Some works have been directly devoted to the study of restricted or (as a more general phenomenon) anomalous diffusion. Such diffusion may be apparent in PGSE NMR as a dependence of the spin-echo attenuation on the diffusion observation time t_d (see Section 3). In two papers, evidence for anomalous diffusion of the protein cytohrome c in aqueous solutions of hyaluronan was reported [87,88]. The anomalous diffusion was evident using other techniques but not directly apparent in the PGSE NMR data.

In a recent work the self-diffusion of PEG confined between lamellae of magnetically aligned bicelles (i.e. disk-shaped aggregates formed by phospholipids) was studied using PGSE NMR [89]. The molecular weights of the PEG samples studied were in the range 200–20,000 Da. The bicelles are oriented with their bilayer normal perpendicular to the static magnetic field, and the PEG chain self-diffusion is therefore measured in the direction (z-direction) perpendicular to the bilayer normal (i.e. parallel to the surfaces of the bilayers). The spin-echo attenuation data show some deviation from a single exponential decay, but this problem was not further addressed in the work. Quoted self-diffusion coefficients D show roughly a scaling according to: $D/D_0 \sim (r_h/H)^{-2/3}$, predicted by scaling theory. D_0 is the corresponding PEG self-diffusion coefficient for the unrestricted chain, r_h is the hydrodynamic radius of the PEG chain, and H is the distance between the oriented bilayers.

4.1.6. Miscellaneous

Finally, in this section, we include some references to studies dealing with PGSE NMR on polymers in solution or polymeric solution systems, that do not fall into the categories listed above. Komlosh and Callaghan observed effects of spin diffusion in stimulated echo PGSE experiments on very high molecular weight polystyrenes in deuterochloroform semidilute solutions (concentrations in the range 5–24 wt%), an effect that was found to be absent in corresponding ordinary Hahn PGSE experiments [90]. The spin diffusion coefficient has to match the coefficient for the translational diffusion of the spins for the effect to be measurable (of the order of 10^{-15} m^2 s^{-1} in the present case).

Dilute solution behaviour of copolymers based on maleic acid has been examined using *inter alia* PGSE NMR in a study by Sauvage et al. [91].

Kriz studied the diffusion of small molecules (cyclohexane, benzene and chloroform) solubilized in aqueous solutions of *inter alia* bilayer and trilayer micelles of poly(hexyl methacrylate)-*b*-(acrylic acid) and poly(2ethylhexyl methacrylate)-*b*-(methyl methacrylate)-*b*-(acrylic acid), respectively [92]. The diffusion behaviour of the solubilizates could be interpreted in all cases as a fast exchange between two sites – bound to micelles, and dissolved in water. No sign of any diffusional barrier between the micellar core and the solvent, even for the trilayer micelles, was evident in this study.

Kriz et al., in two papers on hydrogen bonds of macromolecules, have studied binding of poly(4-vinylpyridine) (PVP) to poly(4-vinylphenol) (PVF) [93,94]. In the first of these papers, binding of tetrahydrofuran (THF) and pyridine and their deuterated analogues to *inter alia* PVF was studied using PGSE proton and deuterium NMR. The binding was quantified using a two-site model for the diffusing molecules, with fast exchange between these two states, regarded as free or bound to the macromolecule:

$$D = (1 - p)D_F + pD_B, \tag{35}$$

where D is the measured diffusion coefficient of the actual component, D_F the corresponding self-diffusion coefficient at the same temperature for the pure liquid (free state), and D_B the self-diffusion coefficient for the molecule in the bound state. These diffusion coefficients may all be measured separately. The value of D_F is obtained by measuring the self-diffusion of the pure solvent, and D_B is identified with the polymer self-diffusion coefficient. This leaves only one unknown, p, which is the degree of binding ($0 \leqslant p \leqslant 1$). In the second paper, the degree of binding of deuterated pyridine was studied in the course of formation of a PVF–PVP complex. By using a combination of several techniques, including theoretical quantum mechanical calculations (based on density functional theory), ^{1}H/^{13}C spin-diffusion measurements, and other methods, it was concluded that there is a high degree of cooperativity in this complex formation [94].

Adsorption of polymer on colloidal quantum dots in (deutero)chloroform solutions, quantified by PGSE measurements, was reported recently by Shen et al. [95]. In this work, ligand exchange between poly(2-(*N*,*N*-dimethylamino)ethyl methacrylate) (PDMA) and trioctylphosphine oxide (TOPO) bound to the surface of CdSe/TOPO quantum dots was demonstrated and quantified by the appearance of the spin-echo attenuation data for the two polymers. A double exponential spin-echo attenuation pattern was observed when the polymer is either bound to the quantum dot surface or exists as free polymer in the solution. This is due to a slow exchange between these two sites in this case. The simultaneous observation of the spin-echo attenuation profiles for the two kinds of polymers clearly showed a displacement of TOPO by PDMA that is preferentially adsorbed to the quantum dot particles. Stimulated echoes were used in these experiments. For the analysis to be quantitative, different spin relaxation times (T_1 and T_2) for the groups of the observed NMR signals used in the PGSE data for polymers in the free and bound states must be taken into account.

4.2. Polymeric melts and melt blends

Polymeric melts are simpler than polymeric solutions in that melts are one-component systems. Polymeric blends may consist of two or more components, that are polymeric in nature, in contrast to polymeric solutions, where at least one component (i.e. the solvent) is a low molecular weight substance. The dynamics in these kinds of system are generally simpler than that of corresponding solutions. Therefore, an interest exists in obtaining high quality data on the self-diffusion of polymer chains, and in testing these data against theoretical models for such systems. As noted above, Nose has previously reviewed PGSE works on *inter alia* polymeric melts up to 1993 [16].

Appel and Fleischer studied the self-diffusion chain length dependence in melts of poly(dimethylsiloxane) (PDMS) and poly(ethylene oxide) (PEO) [96]. Molecular masses of the generally almost monodisperse polymer samples were varied, covering a range up to more than $10M_e$, where M_e denotes the entanglement molar mass. In melts of chains with molecular masses M below M_e, the self-diffusion coefficients D of the highly flexible chains were found to scale with simple Rouse chain dynamics, i.e. $D \propto M^{-1}$. For chains with molecular masses larger than ca. $10M_e$, the value of D scales with M according to reptation dynamics, i.e. $D \propto M^{-2}$. Between these two extremes of the polymer dynamics there is a crossover region where the dependence of D on M is strong, but no simple model for the dynamics can explain the details in this region.

To study the self-diffusion of a macromolecule with a size characterized by some "diameter" d (calculated from the radius of gyration, R_g) the inverse of the product $q = \gamma G\delta$ must exceed this measure of the molecular size. When $(\gamma G\delta)^{-1} < d$, there is a possibility that the observed motion of the spins (protons) is an internal motion of the chains. See for example Ref. [33]. PGSE measurements reported by Appel et al. on melts of rather high molecular weight PDMS, polybutadiene (PB), and poly(isoprene) (PI), were made with very high field-gradients G [97]. The proper self-diffusion coefficient D is time independent, but by using large values of G (of the order of 100 T m^{-1}) and observing large molecules with dimensions exceeding the corresponding q^{-1} one may expect to determine an apparent diffusion coefficient D_{app} that scales with observation time t_d according to $D_{app} \propto t_d^b$, where $b < 0$. Appel et al. determined values for b between -0.57 and -0.73, depending on the kind of polymer. For the PDMS polymer, the crossover from anomalous (time dependent D) to normal diffusion (time independent D) was demonstrated.

In a PGSE and spin–spin (T_2) relaxation study of blends of cyclic and linear PDMS by Cosgrove et al. it was demonstrated that a special entanglement mechanism termed "ring-threading" is operative above a certain molecular weight of the cyclic chains [98a]. For melt blends corresponding to average molecular weight above this molecular weight (ca 3000 Da), the self-diffusion of the components are somewhat lower than the corresponding self-diffusion coefficients for these components in the pure melt state at the same temperature. The experimental temperature was 20 °C [98b].

Rathgeber et al. investigated polymer dynamics in bimodal poly(ethylene) (PE) melts (i.e. binary blends) by a combination of neutron spin-echo spectroscopy and PGSE NMR [99]. A temperature of 236 °C was used in these experiments. The transition from Rouse – to reptation-like diffusion was studied by changing the mass fraction of long polymer chains (with $M \approx 3$ and $4M_e$) in a matrix of shorter chains ($M \approx M_e$). At low mass fractions of long chain polymer (below ca. 50%, depending on the molecular mass of the long chain), there is a deviation from theoretically expected self-diffusion coefficients of these long chains, based on reptation modeling, in that the measured self-diffusion coefficients are lower than those predicted theoretically. The center-of-mass diffusion coefficients measured by neutron spin-echo experiments, on the nanosecond time-scale, are always larger than the corresponding self-diffusion coefficients measured by PGSE NMR (time-scale of milliseconds). Theoretical modeling involving some parametrization of the entanglement effects was shown to account for the experimental data.

Aslanyan et al. have studied macromolecular self-diffusion in binary blends of PEG [100] using pulsed field-gradients up to a maximum of 200 T m^{-1}. From a close analysis of the spin-echo attenuation, also as a function of the diffusion observation time t_d (see Section 3), it was concluded that there is a cluster formation in these systems at the experimental temperature of 70 °C. By studying PEG samples with (number average) molecular weights M_n in the range 2000–40,000 Da (M_w/M_n = 1.1) the following approximate values for the scaling exponent α for the self-diffusion coefficients, $D \propto M_n^{-\alpha}$ (see Eq. (15)), were found: 1.9 for pure melts, 1.2 for a blend with 50% PEG with M_n = 2000 Da, and ~0.9 for corresponding blends with 95% of the low molecular weight PEG.

Fischer et al. have investigated the effect of spin diffusion in melts of PEO, using the fringe-field NMR diffusometry technique [101]. Stimulated spin-echoes of protonated chains were studied in a steady magnetic field-gradient of magnitude 60 T m^{-1} in the stray field of the main magnetic field (9.4 T). It was demonstrated that spin diffusion may dominate the spin-echo attenuation at diffusion observation times longer than ca. 100 ms in the investigated PEO melts, especially for very high PEO molecular weights.

Haley et al. have presented a study on blends of polyisoprene (PI) and poly(vinylethylene) (PVE) [102]. Using PGSE NMR (without Fourier transform of spin-echoes), self-diffusion coefficients for the PI component in blends with low molecular weight perdeuterated PVE were determined as a function of blend composition and temperature in the range 30.5–125.5 °C, well above the glass-transition temperature T_g for both components [102]. The measured self-diffusion coefficients were used to calculate monomeric friction coefficients ζ and terminal relaxation time τ_1, using the Rouse model, that were found to be adequate at least for the PI molecular weights covered. As a function of temperature, ζ and τ_1 were found to fit well to an existing model for the terminal dynamics (the Lodge–McLeish model) in such systems. Other methods like dielectric relaxation and rheology were used together with existing literature polymer dynamics data to test the validity of the Lodge–McLeish model in polymer blends.

von Meerwall et al. have quite recently presented two works on self-diffusion in polyethylene (PE) blends that contain *n*-alkanes as the second component [103,104]. These works are representative of much of the present status of modeling the dynamics of polymer systems in the melted state.

Finally, in this section, we mention that Ozisik et al. have reported a study of nanocomposites consisting of zinc oxide particles in PE melts, covering a range of PE molecular weights, and particle sizes [105]. A combination of *inter alia* NMR spin–spin (T_2) proton relaxation data, probing very local (intrachain) dynamics, and PGSE data shows that, at a constant filler content, the local dynamics of the polymer chains are most affected by small-size particles. However, at the filler loads covered, no obvious effect was observed on the self-diffusion behaviour of the chains, at least for the PE molecular weights covered in that work.

Self-diffusion of ^{129}Xe in the terpolymer ethylene-propylene-diene (EPDM), in semi-crystalline polypropylene (PP), and in a blend of these polymers, was studied using PGSE NMR by Junker and Veeman [106]. The self-diffusion coefficients were found to be of the order of 10^{-12} m^2 s^{-1}, and an order of magnitude higher in EPDM than in PP and in a blend of these. The rate of diffusion of gas in these polymeric materials depends on the pore structures that are present.

4.3. Polymeric gel systems

Polymeric gels may be categorized as soft materials [22]. A percolated polymer network is formed in solvent-containing systems. The networks are viscoelastic with the elastic modulus dominating over the viscous modulus. Such systems find many applications not only because of their mechanical properties, but also because of their value as controlled release agents in medical applications. Polymeric gels may be divided into permanent (chemical) gels or physical (thermoreversible) gels, and an important property is the ability of molecules not forming part of the polymer network to diffuse through the polymer network. Therefore, systems of this kind have attracted, and still attract, a lot of interest also from a fundamental point of view. In principle, field-gradient NMR methods can produce data for all components of such systems: for polymer solutions, various models for the self-diffusion of these components are used in a continuous effort to understand the various mechanisms that are responsible for the observed molecular self-diffusion.

4.3.1. Physical and chemical gels

Physical or thermoreversible gels are typically formed in solutions of polymers that may form intermolecular associations due to, for example, hydrogen-bond forming groups or side chains with lower affinity for the solvent than the main polymer chain. The polymer concentration must be larger than the critical overlap concentration c^*, so that a percolated network of polymer chains may be formed. Protein gels of gelatine may be formed when the temperature is below a certain value. Other gels may be formed above a certain temperature, that in general is dependent on the composition of the gel-forming system. Thermoreversible gel systems respond to a change in temperature in a reversible way. This is due to physical bonds between the polymer chains that are influenced strongly by temperature.

Chemical, or permanent, gels are formed by chemically cross-linking the polymer chains, forming covalent bonds between the chains that can not be broken in a reversible way. So-called hydrogels belong to this sub-group of gel systems.

Walderhaug et al. have investigated gel-forming systems of EHEC dissolved in water in the presence of ionic surfactants [107]. The side chains of these polymers become less hydrated at elevated temperatures, and intermolecular associations between these groups and with surfactants present are thought to provide the gel-forming mechanism. The self-diffusion of the surfactant (sodium dodecyl sulfate, SDS, or hexadecyl trimethylammonium bromide, CTAB) was analyzed according to a two-state model with fast exchange, analogous to Eq. (35). The surfactant molecule is either bound to a polymer molecule with self-diffusion coefficient D_B or is diffusing as a free surfactant molecule in the aqueous solution with a corresponding self-diffusion coefficient D_F. The fraction p of the surfactant molecules that are bound to polymers may thereby be determined. The surfactant concentrations were kept below the critical micelle concentrations, CMC, of the two types of surfactant in the investigated temperature range, so that no free micelles exist. It was demonstrated that p is constant and approximately the same in the sol- and gel states at the same composition of the systems. It was also demonstrated that the cationic surfactant (CTAB), that formed the weakest gels (i.e. with lowest elastic modulus), is bound more strongly to the polymers than the anionic SDS, in that the values for the fraction of surfactant molecules bound to the polymers was found to be higher for CTAB than for SDS at the same concentration of polymer and surfactant. It was also demonstrated that the polymer self-diffusion is measurable in the gel state, even if the spin-echo attenuation is non-exponential and shows a rather complex pattern.

Ray et al. have shown in a study of the self-diffusion of water in a hydrogel of poly(*N*-isopropylacrylamide) (PNIPAM) and over a large temperature range that the water diffusion is only slightly retarded in equilibrium swollen gels compared to corresponding diffusion in pure water [108]. Also, the self-diffusion of the water component may be analyzed according to a two-state model (as in Eq. (35)). The water self-diffusion after reducing the amount of water below the value for the equilibrium swollen state, was found to be considerably lower than in pure water at the same temperature.

Brand et al. recently studied gelation of aqueous solutions of gelatin, and were able to define a gel point from self-diffusion characteristics of the polymer (protein) chains [109]. The minimum of the polymer self-diffusion coefficient as a function of temperature was identified as the gel point. No further discussion on a mechanism describing this phenomenon was given. Hysteresis effects were obvious, in that cooling and heating the solution/gel through such cycles gave different self-diffusion characteristics.

Retama et al. recently published a study of polymer chain dynamics in PNIPAM microgels, where PGSE NMR was used to determine the (cross-linked) polymer self-diffusion [110]. Using quasielastic neutron scattering they determined a diffusive mode with diffusion coefficient similar in magnitude to that of free chains in solution in the swollen state at room temperature. At higher temperatures, in the collapsed state, this diffusion was found to decrease by more than a factor of 10. PGSE measurements on the swollen state revealed that the spin-echo attenuation of the signals from the polymer chains are double exponential decays. The fast component was ascribed to chains in the gel where the concentration of cross-links is low, and the slow component to chains in the gel network where the cross-link density is higher. Deng et al. recently investigated hydrogels of modified cyclodextrins using *inter alia* PGSE NMR [111]. A decrease in self-diffusion coefficient of one of the modified cyclodextrins as a function of concentration was revealed and ascribed to increasing size of the intermolecular complex with this concentration.

Other recent works on solvent and/or polymer self-diffusion in polymer gel systems may be found in Refs. [112–118]. Yamane et al. studied restricted diffusion of solvent (1,4-dioxane) in oriented polypeptide gels, and determined self-diffusion coefficients both along and perpendicular to channels formed by the polypeptide chains [112]. Price et al. used ^{31}P PGSE NMR to study the self-diffusion of phosphine polymers encapsulated in polyurea capsules. Based on the values of the experimentally determined self-diffusion coefficients, it was concluded that the local concentration of phosphine polymer is high, indicating a somewhat swollen capsule wall [115].

4.3.1.1. Probe diffusion studies. Often one adds a molecule, that does not form part of the macromolecular network in the gel, as a probe molecule to investigate various features of the gel. An important property for gel systems is the diffusion of such probes through the gel, as this diffusion may be utilized in, for example, the development of controlled drug delivery applications. Field-gradient NMR studies provide quantization of the self-diffusion of various probe molecules in a gel matrix.

Gomi et al. have studied self-diffusion of water during gelation of rice starch [119]. The gelation takes place at elevated temperature, and by changing the temperature of the samples from room temperature to a temperature at or above the gelation temperature, and subsequent cooling back to room temperature, it was possible to quantify the gelation process. The water self-diffusion is retarded in the gel state as compared to that in the corresponding sol state. By variation of the time during which gelation takes place, it was also possible to determine a rate constant for the gelation of the starch.

The self-diffusion of PEO of a range of molecular weights in aqueous matrices of EHEC/SDS has been investigated in two studies by Walderhaug and Nyström [120] and by Nyden et al. [121]. Obstruction factors, i.e. values of D/D_0 where D is the self-diffusion coefficient of PEO in an EHEC matrix, and D_0 the self-diffusion coefficient of a corresponding water solution of PEO, were determined as a function of EHEC concentration. In the first of these two papers, the gels were physical gels containing SDS [120], while in the other paper the EHEC was chemically cross-linked [121]. In this work a range of molecular weights of rather monodisperse ($M_w/M_n < 1.1$) PEO were examined over a larger EHEC concentration range than used in the former study. The obstruction factors in the sol state of these systems were found to be described, at least partly, by the rather simple relation [121]:

$$D/D_0 = \exp(-\kappa r_H), \tag{36}$$

where r_H is the hydrodynamic radius of the probe molecules (see also above), and κ is the inverse screening length of the polymer matrix. In the gel state (1 wt% EHEC), the echo attenuation of the PEO chains were generally non-exponential. The MSD ($\langle Z^2 \rangle$) was evaluated from these data, and seemed to scale with molecular weight M according to Rouse-type dynamics (see above). A controversy at the time these studies were carried out was whether the probe diffusion was Fickian or not. In the physical gels, the PEO spin-echo data, as a function of diffusion observation time, can be interpreted as showing a deviation from simple Fickian diffusion. In the chemical gels no departure from Fickian diffusion was found, however. One point is that the two kinds of systems are not identical, and in both studies the attenuation of the echoes from the PEO chains were found to be more complex in the gel state than in the corresponding sol state.

Annaka et al. investigated PEG chain diffusion in two types of hydrogels based on *N*-isopropylacrylamide (NIPA) [122]. The two types of gels were so-called normal type NIPA gels and comb-type grafted NIPA gels, with a smaller matrix mesh size than in the former case. The PEG was of rather low molecular weight (1000 Da). From simple mono-exponential spin-echo-decays the self-diffusion coefficients D of the PEG chains in these two kinds of gels were easily determined, and interpreted according to $D/D_0 = \exp(-r_h/\xi)$ (see Eq. (36)), where ξ denotes the mesh size of the polymer network, and D_0 is the corresponding self-diffusion coefficient of the PEG chains in absence of the polymer matrix. At low cross-linking density ξ may be expected from scaling theory to depend on polymer concentration ϕ according to: $\xi \propto \phi^{-3/4}$. For the normal NIPA hydrogels, the self-diffusion of the PEG chains and also that of the water were found to comply with these predictions, while for the comb-type grafted gels where the density of cross-links is higher, the self-diffusion coefficients of the PEG chains are considerably lower than predicted in this way.

Yamane et al. investigated the self-diffusion of amino acids in so-called Merrifield network polystyrene gels [123,124]. These gels are resin type gels where the PS chains have been cross-linked using divinylbenzene. The beads were then swollen using (deuterated) *N*,*N*-dimethylformamide (DMF) containing a dissolved amino acid. The self-diffusion coefficients for *tert*-butyloxycarbonyl-L-phenylalanine (Boc-Phe), and, in the first of these papers, also *tert*-butyloxycarbonyl-glycine (Boc-Gly) were determined. Temperature dependence of the self-diffusion coefficients were used to obtain activation energies from Arrhenius plots, as a function of gel composition. An interesting time-dependence of the self-diffusion coefficients D for the last-mentioned component was reported, in that D was shown to be dependent on the diffusion observation time Δ, in the range 5–50 ms, an effect ascribed to restricted diffusion. Investigating also cross-linked ethoxylate acrylate gel with DMF as solvent, Boc-Phe and PEO ($M_w = 1500$ Da) spin-echo attenuation data were found to be double exponential at short observation times, and became mono-exponential at longer values of Δ. The effects were ascribed to inhomogeneities in the gel matrix [124].

Kwak and Lafleur investigated self-diffusion of macromolecules and micelles in gels based on the polysaccharides curdlan [125] and dextran [126]. In the first of these papers the self-diffusion data were shown to comply with models for diffusion of probe molecules in polymer matrices, developed by Kukier and Petit. These are phenomenological models. They are therefore not directly predictive in nature, but contain explicit dependence of the probe molecule self-diffusion coefficients on polymer concentration. Temperature dependence of the diffusion coefficients over the investigated temperature range were shown to form linear Arrhenius plots and used to calculate corresponding activation energies E_a. A small increase in E_a with polymer concentration was demonstrated [125]. In the second work, dealing with self-diffusion of small-size probe molecules and surfactants in dextran solutions and gels it was reported that especially the nonionic Triton X-100 surfactant micelles are retarded to a rather high degree compared to the self-diffusion of the micelles in the absence of polymer [126]. This effect is most probably related to the adsorption of the surfactant micelles to the polymer chains.

Rosen et al. determined the self-diffusion of the anionic surfactant SDS in aqueous solutions and cross-linked gels of EHEC [127]. In this study, the SDS concentration was kept below the CMC of the surfactant, so that no free micelles were present in the investigated systems. A high degree of surfactant binding, as analyzed according to Eq. (35) was demonstrated. In the gel state the spin-echo attenuation of the surfactant was found to be non-exponential, and analyses of the echo attenuations were performed, based on an assumption of a distribution of diffusivities according to Eq. (18). The distributions that were determined in this way were dependent on the diffusion observation time, however, showing a rather complex dynamics for the probe molecules.

Baldursdottir and coworkers have reported a study of PEO chains in aqueous solutions and gels based on alginate, containing also the photosensitizer riboflavin [128]. The spin-echo attenuation of the PEO chains in general show some deviation from single exponential character, and were analyzed using Eqs. (22) and (23). In a qualitative way, the self-diffusion data of the probes show that irradiating these systems the probe molecules are less obstructed, due to scission processes in the polymer matrix.

Therien-Aubin and coworkers recently reported a PGSE study of *inter alia* oligomers of ethylene glycol, methyl ester derivatives of these, and acetic acid in (aqueous) gels of polymer matrices containing groups that form hydrogen bonds to the probes, such as poly(vinylalcohol) and poly(acrylic acid) [129]. Both the Petit [76] and the Phillies diffusion models [135] (see also the last paragraph of this sub-section) could be fitted to the probe self-diffusion data as a function of polymer matrix concentration. The combination acetic acid and poly(acrylic acid) gives rise to ionic interactions between probes and matrix. From the temperature dependence of the probe self-diffusion coefficients, the activation energies for the self-diffusion of the probes in the polymer matrices were determined, and were shown to be fairly constant for the various combinations of probes and matrices.

Ferrero and coworkers also recently presented a study on self-diffusion of salicylate in hydrogels of cellulose ethers [117]. The probe self-diffusion data were found to fit to free-volume theory predictions over the matrix concentrations investigated. In principle, this also was found to hold for the water self-diffusion, but the dependence on the polymer matrix concentration is much less in this case, and the water behaves more like bulk water from a diffusion point of view, even in rather concentrated gels.

Other, recent studies on probe diffusion in polymer gel systems have been reported [130–137]. In Refs. [130–133], PEG of various molecular weight were used as probe molecules in the gel network. Obstruction and restricted diffusion effects were observed and correlated with probe molecule size and structure (such as the mesh size) of the gel network.

Kamiguchi and coworkers used polystyrenes of various molecular weight as probe molecules in a PMMA gel and also found evidence of restricted probe self-diffusion, especially for higher molecular weight probes, that was correlated with the gel structure [134]. Modesti and coworkers in a recent work compared the behaviour of small solvent molecules and dye probes in octane-swollen poly(dimethylsiloxane) solutions and (cross-linked) gels, and compared the self-diffusion data to predictions based on free-volume theory and the hydrodynamic theory of Phillies [135]. Ref. [136] reports a study of dendrimer diffusion in κ-carrageenan gels. Salt conditions influence the gel structure, and by using dendrimers of different sizes it was possible to study the influence of the gel structure on the self-diffusion of the probe molecules. Gagnon and Lafleur have measured self-diffusion of various phosphates in curdlan hydrogels using ^{31}P PGSE NMR. Obstruction effects were rationalized within a model suggested by Ogston [137].

4.3.2. Gel electrolytes

Gel electrolytes consist of low molecular weight electrolytes dissolved in a gel matrix formed by cross-linked nonionic polymers based on PEO, polyacrylonitrile (PAN) or other polymers that are soluble in polar solvents. The matrix forms a mechanically stable device, holding the electrolyte in place, and is used in lithium battery or membrane technologies. Important properties in such systems are the transport and conductivity of ions. Self-diffusion of Li^+, fluorine-containing ions, and proton-containing ions, using mainly ^{7}Li, ^{19}F and ^{1}H PGSE measurements, have been determined for such systems and reported in a range of papers during the last decade or so [138–158]. In general, the self-diffusion coefficients of the ions are easily accessible, since the spin-echo attenuation is found to be single exponential or close to single exponential. The self-diffusion coefficients of the ions are in general connected with the electrolyte's conductivity σ according to the Nernst–Einstein equation (assuming univalent ions):

$$\sigma = \frac{Ne^2}{k_B T}(D_+ + D_-) \quad (37)$$

where D_+ and D_- are the self-diffusion coefficients for the cations and the anions, respectively, N denotes the number of cations (or anions) per unit volume, and e is the elementary charge. The other symbols have their usual meaning. Eq. (37) does not take into account ion association which may be taken care of by introducing a factor of $(1 - \zeta)$, with $0 \leqslant \zeta < 1$ in the equation [139,141]. However, exceptions to Nernst–Einstein behaviour has been demonstrated and pointed out [145].

The relative importance of the cations and the anions for the conductivity in such systems is directly accessible from the measurements of the self-diffusion of the ions, and transference numbers τ for cations and anions may easily be calculated from the diffusion coefficients, by simple relations: $\tau^+ = D_+/(D_+ + D_-)$; $\tau^- = D_-/(D_+ + D_-)$. See, for example, Ref. [146]. Activation energies of the individual ions may also be determined by PGSE measurements at various temperatures [140,141].

Proton conducting membranes are used in fuel cell technology. In methanol fuel cells such membranes made of polymers (ionomers) that may contain fillers (i.e. composite membranes) may be utilized. As a few examples of recent self-diffusion studies of water and possible fuels in these membranes, we cite the following Refs. [159–162]. Water self-diffusion coefficients may be used to calculate proton conductivity using the Nernst–Einstein equation (Eq. (37)). The proton conduction mechanism may to some extent be explored by comparing the calculated (proton conductivity) with experimental data [161]. Fuel self-diffusion is a measure of the degree of so-called fuel crossover, in that a high self-diffusion coefficient and, as a consequence, a high crossover of the fuel (e.g. methanol) may result in a reduced efficiency of the cell. Increasing the knowledge of the effect of the chemical and physical structure of the polymer membrane on the diffusional behaviour of the fuel may make it possible to optimize the membranes for use in cells based on various fuels. See e.g. Ref. [160].

4.3.3. Liquid crystalline systems

This topic represents a very active field of research, and there are many scientific papers on measurement of self-diffusion of various components in such systems. We restrict ourselves here to contributions from the last five years. Again, the cited literature is not meant to form a complete and exhaustive list, but should hopefully provide an account of the type of information that is presently being sought, and the state of the art of NMR equipment that is used to obtain this information [163–174]. Anisotropic self-diffusion of small molecules such as ethane or water along and perpendicular to the mesophase director in macroscopically oriented liquid crystalline samples has been demonstrated in Refs. [164,165], respectively. Surfactant self-diffusion has also been studied in Refs. [163,168,169]. Water self-diffusion has been studied where water binding to surfactant and obstruction effects (see above) are observed [173], and where the water self-diffusion in powder samples (i.e. samples of randomly oriented liquid crystal directors) has been determined [171]. In a very recent paper, Li and coworkers report self-diffusion data on ionic liquids in a polyelectrolyte membrane (Nafion), obtained by a combination of proton and ^{19}F PGSE measurements of both cations and anions [174]. The influence of the amount of water present on the diffusion of the ions was studied, and interesting hysteresis effects were demonstrated during heating/cooling cycles. Finally, Åslund and coworkers have suggested a method to determine the size of anisotropic domains in lyotropic liquid crystalline systems, by a systematic variation of the various experimental parameters in a PGSE experiment, thereby observing deviation from a Gaussian process, i.e. anomalous diffusion processes [172].

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